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Part I

Time Series

Chapter 1

Introduction

1.1 Informal introduction

Definition 1.1.1 (Informal)**.** A time series is a collection of observations $\{x_1, x_2, \dots, x_n\}$ on a given phenomenon, ordered in time ($t = 1, \dots, n$).

Example 1.1.1. Hourly temperature of a patient subject to a treatment, daily price of an asset return, monthly index of industrial production, quarterly U.S. GDP, annual immigrations in Italy.

Remark 1. There are two ways of tackling time series analysis:

1. **decomposition** (classical time series analysis, 1930's): what we observe is the combination of some unobserverd components we can estimate making some assumption.
   Models can be unobserved component models, state space methods or non-parametric (filtering and smoothing methods, local polynomial regression, kernel estimation);

2. **modelling** (modern time series analysis, 1970's, Box Jenkins): we specify a hypothetical probability model to represent the data (which describes the dependency of observation in time). After an appropriate family of models has been chosen, it is then possible to estimate parameters, check goodness of fit etc.
   Model can be linear (ARMA and generalization) or non-linear (GARCH, TAR, ARFIMA, GAS)[1]

Remark 2. Some remarks:

- these observations come from random variables that are likely to be *dependent*;

- *frequency of observation* and *length of the series*/number of observation depends on nature of the phenomenon (eg it doesn't make sense to measure temperature of patient yearly); this typically leads to using different methods for different kind of series

[1] In un corso base si fanno le decomposizioni classiche e la modellazione lineare, la modellazione non lineare è advanced

- aims of ts analysis is to analyse the dependence within time series (univariate) or between series (multivariate), and/or to produce nowcasts (real time predictions) and forecasts

Remark 3. First steps

1. Plot the series and describe the main features of the graph, checking in particular whether is there:

   - an increasing/decreasing trend,
   - a seasonal component/a cyclical behaviour (if so the periods seems to be constant or varying?)
   - any apparent sharp changes in behavior/discontinuities in the series: such as a sudden change of level, it may be advisable to analyze the series by first breaking it into homogeneous segments;
   - any outlying observations: they should be studied carefully to check whether there is any justification for discarding them (as for example if an observation has been incorrectly recorded)

   Inspection of a graph may also suggest the possibility of representing the data as a realization of the process (the classical decomposition model)

   $$X_t = m_t + s_t + \varepsilon_t \tag{1.1}$$

   with

   - m_t is a slowly changing function known as a **trend component**,
   - s_t is a function with known period d referred to as a **seasonal component**
   - ε_t a random **noise component** that is stationary

   The model could be even specified by pre-logging the observed variable in order to obtain a relation which is multiplicative in the three component

   $$\log X_t = m_t + s_t + \varepsilon_t$$

2. one typically does some **transformation** in order to prepare the sequence to be analyzed. The idea is to remove the trend and seasonal components (ITSF 1.5) to get stationary residuals (sec 1.4 ITSF).
   If seasonal and noise fluctuations appear to *increase* with the level of the process, then a preliminary transformation of the data (eg log) is often used to make the transformed data more compatible with model 1.1.

3. Choose a model to fit the residuals, making use of various sample statistics including the sample autocorrelation

4. Forecasting will be achieved by forecasting the residuals and then inverting the transformations described above to arrive at forecasts of the original series

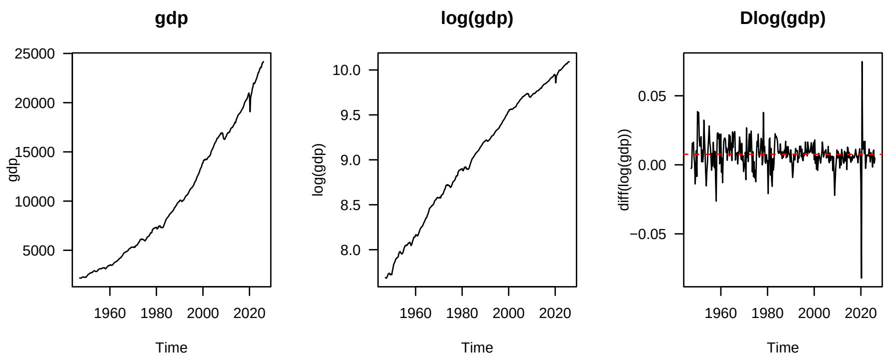


Figure 1.1: US GDP (1947-2026)

1.2 Data examples

Example 1.2.1. eg the gdp follow an exponential trend, but if we take the logarithm the increase become linear and if we further take the differences of the logarithm of the series we have an estimate of the *growth rate*[2] which become approximately constant in time. So the growth rate is stationary in mean; there seems to two main phases (the first up to 1980 with greater variability, then the following more stable but a spike due to covid) so we could think of two different generating process

```
library(lbts)
head(gdp)

##      Qtr1 Qtr2 Qtr3 Qtr4
## 1947 2183 2177 2172 2206
## 1948 2240 2277

par(mfrow = c(1,3))
plot(gdp, main = "gdp", las=1)
plot(log(gdp), main ="log(gdp)",las=1)
plot(diff(log(gdp)), main = "Dlog(gdp)", las=1)
abline(h=mean(diff(log(gdp))), lty="dashed", col="red")
```

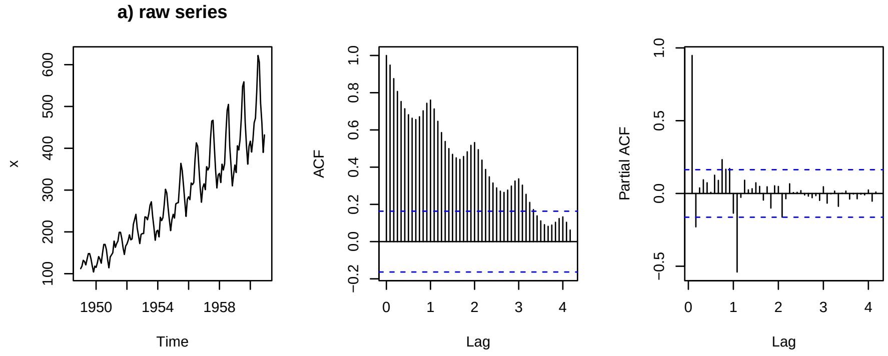
For plotting

- a raw R vector as time series or multiple time series in the same graph use `ts.plot`
- for a single time series object standard `plot.ts` is better programmable

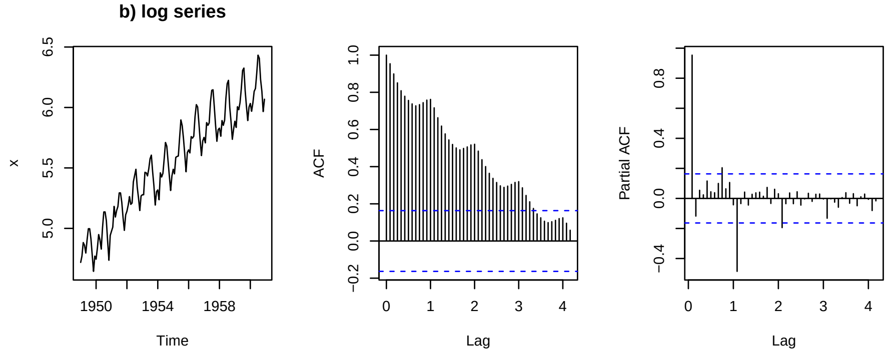
Example 1.2.2.
```
library(TSA)
options(width=90)
```

[2]This is discrete equivalent of derivative (and we know that if we have a linear increasing function and take the derivative it becomes a constant)

```r
h <- function(x) round(head(x, n=20), 3)

## a) Raw serie (monthly data)
x <- AirPassengers
h(x)

##      Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
## 1949 112 118 132 129 121 135 148 148 136 119 104 118
## 1950 115 126 141 135 125 149 170 170

ts_plot(x, main="a) raw series")
```

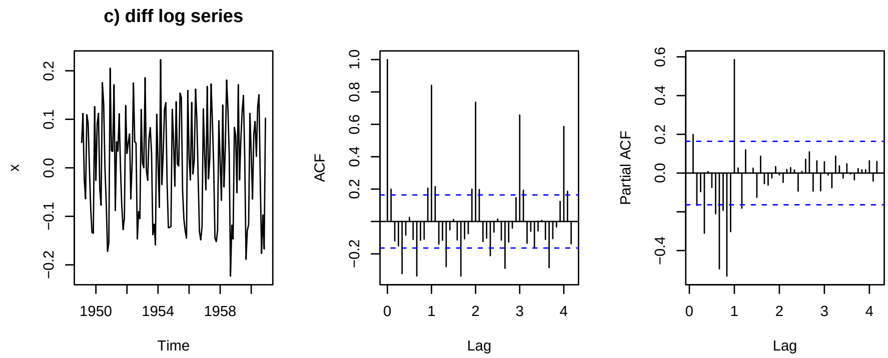



```r
## b) Log series
h(log(x))

##        Jan   Feb   Mar   Apr   May   Jun   Jul   Aug   Sep   Oct   Nov   Dec
## 1949 4.718 4.771 4.883 4.860 4.796 4.905 4.997 4.997 4.913 4.779 4.644 4.771
## 1950 4.745 4.836 4.949 4.905 4.828 5.004 5.136 5.136

ts_plot(log(x), main="b) log series")
```

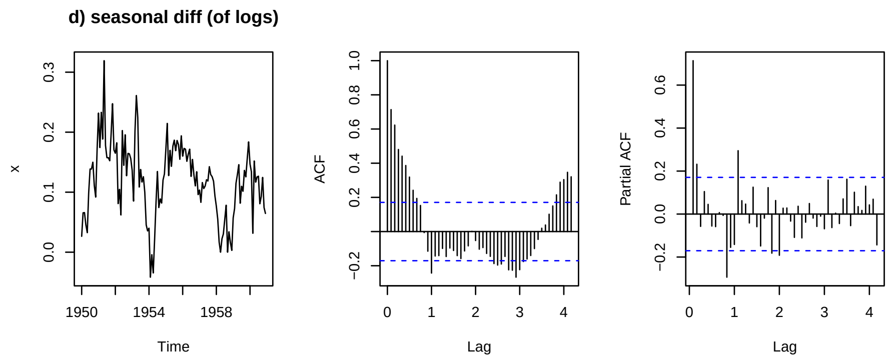



```r
## c) diff log series
h(diff(log(x)))
```

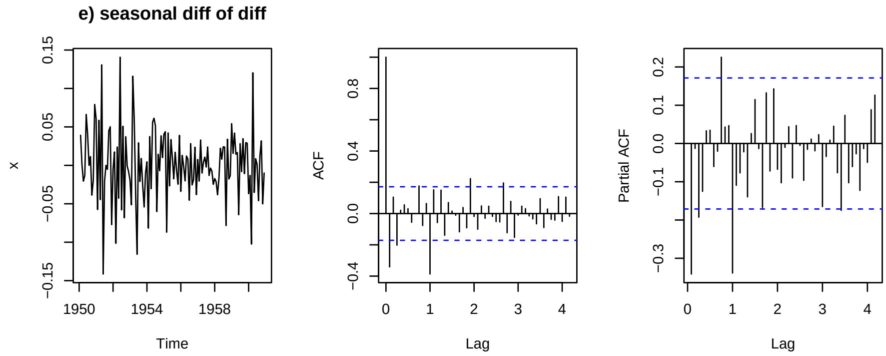
```
##          Jan    Feb    Mar    Apr    May    Jun    Jul    Aug    Sep    Oct    Nov    Dec
## 1949          0.052  0.112 -0.023 -0.064  0.109  0.092  0.000 -0.085 -0.134 -0.135  0.126
## 1950 -0.026  0.091  0.112 -0.043 -0.077  0.176  0.132  0.000 -0.073

ts_plot(diff(log(x)), main = "c) diff log series")
```



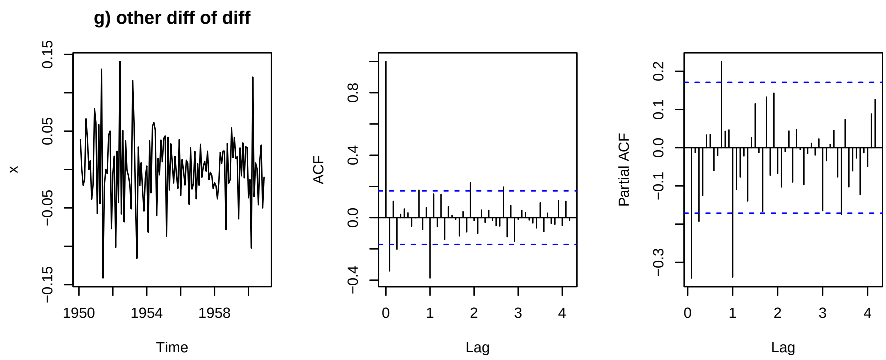

```
## d) seasonal difference (of logs)
h(diff(log(x), 12))

##        Jan   Feb   Mar   Apr   May   Jun   Jul   Aug   Sep   Oct   Nov   Dec
## 1950 0.026 0.066 0.066 0.045 0.033 0.099 0.139 0.139 0.150 0.111 0.092 0.171
## 1951 0.232 0.174 0.233 0.188 0.319 0.178 0.158 0.158

ts_plot(diff(log(x), 12), main = "d) seasonal diff (of logs)")
```




```
## e) seasonal differences of the differences
y <- diff( diff(log(x)), 12)
h(y)

##          Jan    Feb    Mar    Apr    May    Jun    Jul    Aug    Sep    Oct    Nov    Dec
## 1950          0.039  0.000 -0.020 -0.013  0.066  0.040  0.000  0.011 -0.039 -0.019  0.079
## 1951  0.061 -0.057  0.059 -0.045  0.131 -0.141 -0.020  0.000 -0.005

ts_plot(y, main="e) seasonal diff of diff")
```




```
# g) the other diff in diff
y <- diff( diff(log(x), 12)  )
h(y)

##           Jan     Feb     Mar     Apr     May     Jun     Jul     Aug     Sep     Oct     Nov
## 1950            0.039   0.000  -0.020  -0.013   0.066   0.040   0.000   0.011  -0.039  -0.019
## 1951   0.061  -0.057   0.059  -0.045   0.131  -0.141  -0.020   0.000  -0.005

ts_plot(y, main="g) other diff of diff")
```



1.3 First definitions

Remark 4. Formally a time series is a single realisation of a stochastic process.

Definition 1.3.1 (Time series)**.** A time series $(x_1, \ldots, x_n)$, with $n < \infty$ [3] is *one, finite,* realisation from a stochastic process $(X_t)_{t \in T}$, where the single random variable are $X_t : (\Omega, \mathscr{A}, \mathbb{P}) \to (\mathbb{R}, \beta(\mathbb{R}), \mathbb{P}_X)$.

Example 1.3.1. Prof uses $T = \mathbb{Z}$ so we have a huge, infinity, number of random variales $\{\ldots, X_{t-1}, X_t, X_{t+1}, \}$, of which we observe only a *single* realization for

[3] In general T is a numeric set of times: if finite or countable we have *discrete-time* time series, if continuous we have *continuous-time* time series. Here as in ITSF, focus seems more on discrete-time time series.

a *small subsequent subset* of the random variables coming from the random process eg $(x_t, x_{t+1}, \dots, x_{t+n})$.

Definition 1.3.2 (Law of stochastic process)**.** It's the family of multivariate distribution generated by the process $\{F_{\mathbf{t}}(\mathbf{x})\}_{\mathbf{t}\in\mathcal{T}}$ such that for $n = 1, 2, \dots$

NB: Should be the finite dimensional distributions in Rigo's terms.

$$F_{\mathbf{t}}(\mathbf{x}) = \mathbb{P}\left(X_{t_1} \leq x_1, \dots, X_{t_n} \leq x_n, \right)$$

where $\mathcal{T} = \{\mathbf{t} = (t_1, \dots, t_n) \in T^n, t_1 < t_2 < \dots < t_n\}$ and $\mathbf{x} = (x_1, \dots, x_n)$.

Remark 5. So it's the collection of all the univariate cumulative distributions at each time, the bivariate distribution etc.

Important remark 1 (Problem with ts). Some remarks:

- our aim is describing the dependence among the random variables in the process: to get information on the stochastic process that generated the time series we see.
- we would reach this perfectly if we could access the law above, but often this is not known (even if we can prove its existence, via Kolmogorov or Tulcea/Ionescu) or too difficult to be obtained (too many parameters to be estimated from the limited available data). That is, based on one single realization (and observed for limited amount of time) it's impossible to describe the data generating process (unless we make further assumptions).
- Thus we often *restrict* the class of stochastic process investigated to those whose stochastic dependence can be described to something we can observe, that is moments (*unconditional* and *conditional*); first and second order moments are of particular interest and we focus on properties of sequence of random variables depending only on these (*second-order properties*).
- to guarantee the validity of inference based on moments (and limited observation) however, process has to possess some homogeneity conditions, such as *stationarity*[4], *invertibility* and *ergodicity*
- we may lose a certain amount of information by looking at second-order properties of random process but as in chapter 2 of ITSF theory of minimum MSE linear predictor depends only on the second-order properties, thus providing further justification for the use of the second-order characterization of time series models.[5].

Definition 1.3.3 (Moments of a stochastic process)**.** First and second moments of $\{X_t\}_{t\in T}$ are

$$\begin{aligned}
\mu_X(t) = \mu_t = \mathbb{E}\left[X_t\right] &= \int_{\mathbb{R}} x_t \, \mathrm{d}F_(X_t)(x) \\
\gamma_X(0,t) = \gamma_{0,t} = \mathrm{Var}\left[X_t\right] &= \mathbb{E}\left[(X_t - \mu_t)^2\right] \\
\gamma_X(r,s) = \gamma_{r,s} = \mathrm{Cov}\left(X_r, X_s\right) &= \mathbb{E}\left[(X_r - \mu_r)(X_s - \mu_s)\right] \\
&= \mathbb{E}\left[X_r X_s\right] - \mathbb{E}\left[X_r\right]\mathbb{E}\left[X_s\right]
\end{aligned}$$

[4] Intended weak stationarity, which we will assume for all the course except here and there where stronger assumption is needed

[5] Btw/Furthermore when all the joint distributions are multivariate normal, the second order properties of the process completely determine the joint distribution and thus thive a complete probabilistic characterization of the sequence of rvs.

with the last called *autocovariance function (ACF)*.

Definition 1.3.4 (Strong/strict stationarity)**.** A process if joint distributions does not change for shift of times, that is

$$(X_{t_1}, \dots, X_{t_k}) \sim (X_{t_1+h}, \dots, X_{t_k+h})$$

for all $k \geq 1$, and for all $t_1, \dots, t_h$, $h \in \mathbb{Z}$. Or in other terms

$$F_{\mathbf{t}}(\mathbf{x}) = F_{\mathbf{t}+\delta}(\mathbf{x}), \quad \forall \delta$$

NB: If in general we speak about stationarity, weakly is intended (ITSF p. 13)

Definition 1.3.5 (Weakly stationarity)**.** A process is weakly-stationary if the *first and second order moments exist* (finite) and are constant/do not depend on the time. In this case the moments:

$$\begin{aligned}
\mu_X = \mu = \mathbb{E}\left[X_t\right] < \infty, \quad \forall t \in T \\
\gamma_X(0) = \gamma(0) = \gamma_0 = \mathrm{Var}\left[X_t\right] < \infty, \quad \forall t \in T \\
\gamma_X(k) = \gamma(k) = \gamma_k = \mathrm{Cov}\left(X_t, X_{t+k}\right) < \infty, \quad \forall t \in T, k = 0, \pm 1, \pm 2, \dots
\end{aligned}$$

The first two above are constant while the third, called autocovariance function, does not depend on the time chosen (for one of the two variables, say the first) but only on the distance $k = |t - s|$ between the variables. Furthermore by properties of covariance $\gamma_{-k} = \mathrm{Cov}\left(X_{t+k}, X_t\right) = \mathrm{Cov}\left(X_t, X_{t+k}\right) = \gamma_k$.

Remark 6 (Autocovariance function importance)*.* So in stationary process we may have different realizations and spaghetti like movement but the vertical means is the same, as well as vertical spread in time; since mean and variance for these process is constant we look at *autocovariance function as key tool* to say something on process.
Auto means covariance of the process with self; we call it *function* because at each $k = 0, \pm 1, \pm 2, \dots$ we have an estimate which we can plot (stick function). Since the function $\gamma_k = \gamma_{-k}$ we tipically plot only the positive side (first and fourth quadrant) and looking at it we can try to tell which model generated the observed values

Proposition 1.3.1 (Covariance useful property)**.** *If* $\mathbb{E}\left[X^2\right] < \infty, \mathbb{E}\left[Y^2\right] < \infty, \mathbb{E}\left[Z^2\right] < \infty$ *and* $a, b, c \in \mathbb{R}$

$$\mathrm{Cov}\left(aX + bY + c, Z\right) = a\,\mathrm{Cov}\left(X, Z\right) + b\,\mathrm{Cov}\left(Y, Z\right) \tag{1.2}$$

Definition 1.3.6 (Autocorrelation function)**.** For a stationary stochastic process, it's an adimensional version of the ACV, defined as

$$\rho_X(k) = \rho_k = \frac{\gamma_k}{\gamma_0}, \quad k = 0, \pm 1, \pm 2$$

with $|\rho_k| \leq 1$ [6].

Important remark 2*.* We have that:

- weak stationarity $\not\Longrightarrow$ strong stationarity (obviously, as moment conditions do not imply, except for densities completely characterised by the first two moments, like the Gaussian, distributional condition)

[6] Due to Cauchy-Schwartz inequality where we define the inner product $\langle X, Y \rangle$ like the expectation $\mathbb{E}\left[XY\right]$

- if $\{X_t\}$ is strictly/strong stationary AND first and second moments are finite ($\mathbb{E}\left[X_t^2\right] < \infty$, $\forall t$), then $\{X_t\}$ is also weakly stationary

Important remark 3 (Stationarity and ACV/ACF). It is not required that the process is stationary in order for autocovariance and autocorrelation to be defined but tipically they have sense for stationary processes. Indeed if the process is non stationary the autocovariance function does not give us that much information on the underlined data generating process.

1.4 Examples

Remark 7. In this section we see examples of stochastic processes and their autocorrelation function.

1.4.1 White noise

Example 1.4.1 (White noise)**.** $\varepsilon_t \sim \mathrm{WN}\left(0, \sigma^2\right)$ (read ε_t is a white noise process 0 σ^2) if composed by random random variables with uncorrelated zero mean σ^2 variance random variables: $\forall t \in \mathbb{Z}$

$$
\begin{aligned}
&\mathbb{E}\left[\varepsilon_t\right] = 0 \\
&\mathrm{Var}\left[\varepsilon_t\right] = \mathbb{E}\left[\varepsilon_t^2\right] = \sigma^2 \\
&\mathrm{Cov}\left(\varepsilon_s, \varepsilon_t\right) = \mathbb{E}\left[\varepsilon_s \varepsilon_t\right] = 0, \quad \forall t \neq s
\end{aligned}
$$

WN is a weakly stationary process, since fullfill requirements of definition (being the three above finite $\forall t \in \mathbb{Z}$).

1.4.2 IID noise

Example 1.4.2 (IID noise)**.** Differently from the previous not only variables are *uncorrelated* but even independent and identically distributed. In IID noise random variables of the process are *independent and identically distributed with zero mean*:

$$(X_1, \ldots, X_n), \quad X_t \sim \mathrm{IID}\left(0, \sigma^2\right)$$

Thus

- $\mathrm{IID}\left(0, \sigma^2\right)$ is a $\mathrm{WN}\left(0, \sigma^2\right)$ but the converse does not hold (independence implies zero correlation but not the converse);
- any finite dimensional distribution is just the product of the marginals of each time (with F being the common cumulative distribution function):

$$
\begin{aligned}
\mathbb{P}\left(X_1 \leq x_1, \ldots, X_n \leq x_n\right) &= \mathbb{P}\left(X_1 \leq x_1\right) \cdot \ldots \cdot \mathbb{P}\left(X_n \leq x_n\right) \\
&= F(x_1) \cdot \ldots F(x_n)
\end{aligned}
$$

- there's no dependence between observations and knowledge of $X_1, \ldots, X_n$ is of no value for predicting the behaviour of X_{n+h}

$$\mathbb{P}\left(X_{n+h} \leq x | X_1 = x_1, \ldots, X_n = x_n\right) = \mathbb{P}\left(X_{n+h} \leq x\right), \quad \forall h \geq 1$$

It's

- a simple model for a time series (with no trend or seasonal component);
- an important as building block for more complicated models;
- weakly stationary (if $\sigma^2 < \infty$) since $\forall t \in \mathbb{Z}$:

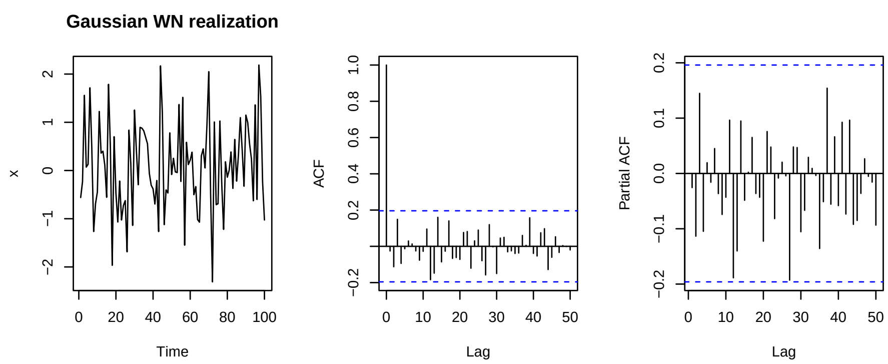
$$\mathbb{E}\left[X_t\right] = 0$$
$$\operatorname{Var}\left[X_t\right] = \mathbb{E}\left[X_t^2\right] = \sigma^2$$
$$\operatorname{Cov}\left(X_t, X_{t+h}\right) = \begin{cases} \sigma^2 & \text{if } h = 0 \\ 0 & \text{if } h \neq 0 \end{cases}$$

Not necessarily an IID process is weakly stationary: eg $\varepsilon_t \sim \mathrm{T}\left(\nu = 1\right)$ is IID process but it's not stationary because has second moment infinite.

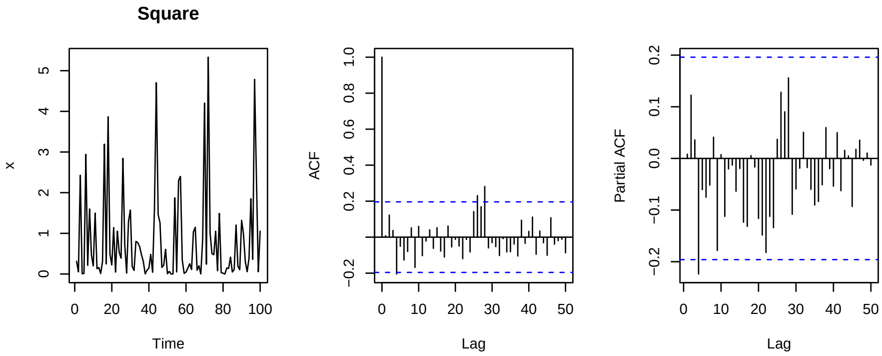
Example 1.4.3 (Gaussian white noise ($NID(0,1)$))**.** As example of IID RP gaussian white noise (means that the process is actually IID Gaussian), where $\varepsilon_t \sim \mathrm{N}\left(0,1\right)$ are iid normal;

- process is weakly stationary because $\mathbb{E}\left[\varepsilon_t\right] = 0$, $\operatorname{Var}\left[\varepsilon_t\right] = 1$ and $\operatorname{Cov}\left(\varepsilon_t, \varepsilon_s\right) = 0$ (due to independence)
- this normal IID process is also *strongly stationary.*

```r
set.seed(123)
y <- rnorm(100) # Gaussian White Noise process simulation
ts_plot(y, main="Gaussian WN realization")
```

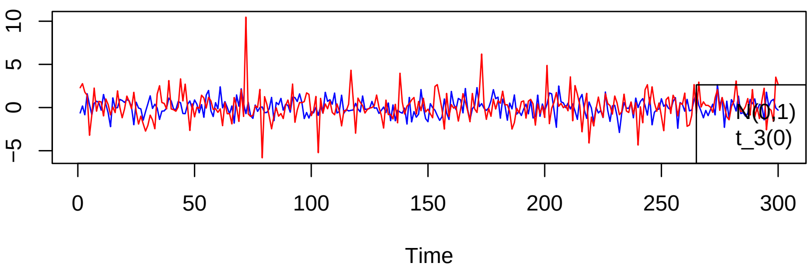



```r
# non linear transformations of the data (eg squares) do not change the
# autocorrelation of the series (as it is an IID series)
ts_plot(y^2, main="Square")
```



Example 1.4.4. Other simulations: T vs normal white noise

```
n <- 300

## Gaussian and sudent t white noise processes
set.seed(1)
wnn <- rnorm(n=n)
wnt <- rt(n=n, df=3) # with lower df the process have larger
# noise/variability and extreme observation

mts_plot(cbind(wnn, wnt), col = c("blue", "red"))
legend("bottomright", legend=c("N(0,1)", "t_3(0)"))
```



1.4.3 Random walk

Example 1.4.5 (Random walk)**.** Defined as

$$(S_0, S_1, S_2, \dots)$$

with

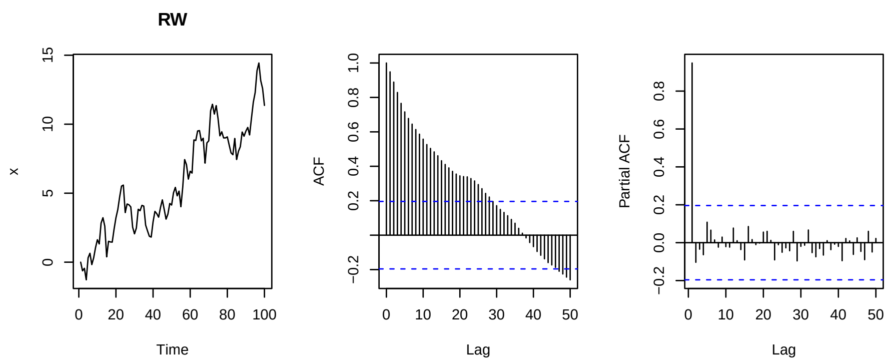
$$S_n = X_0 + \sum_{t=1}^{n} X_t$$

where $X_0 \perp\!\!\!\perp X_t$, X_t are iid, $\forall t \geq 1$. For example if $X_0 = 0$ and $X_t \sim \text{IID}\left(0, \sigma^2\right)$ we obtain a random walk with zero mean (that is $\mathbb{E}\left[S_n\right] = 0, \forall n$).
It's *not stationary*: eg if we consider the zero mean random walk

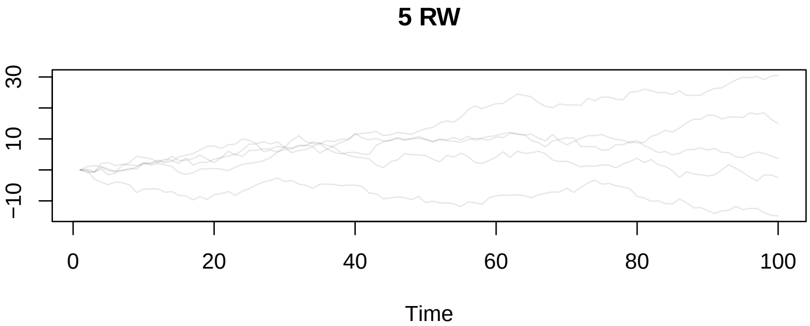
$$\begin{aligned}
\mathbb{E}\left[S_t\right] &= 0 \\
\operatorname{Var}\left[S_t\right] &= \mathbb{E}\left[S_t^2\right] - \left[\mathbb{E}\left[S_t\right]\right]^2 = \mathbb{E}\left[\left(\sum_{i=1}^{t} X_i\right)^2\right] \overset{(\perp\!\!\!\perp)}{=} \sum_{i=1}^{t} \mathbb{E}\left[X_i^2\right] = t\sigma^2 \\
\operatorname{Cov}\left(S_{t+h}, S_t\right) &= \operatorname{Cov}\left(S_t + X_{t+1} + \ldots X_{t+h}, S_t\right) \overset{(1)}{=} \operatorname{Cov}\left(S_t, S_t\right) = t\sigma^2
\end{aligned}$$

where (1) by applying 1.2.
Thus both second moment and covariance, depending on t, may not be finite.

```
sim_rw

## function (n = 100, x0 = 0, xt = rnorm)
## {
##     x <- x0
##     xts <- xt(n = n)
##     rw <- cumsum(c(x0, xts))
##     rw[-length(rw)]
## }
## <bytecode: 0x59bc3823ff60>
## <environment: namespace:lbts>

set.seed(1)
x <- sim_rw()
ts_plot(x, main="RW")
```




```
# multiple ts
set.seed(2)
mts_plot(replicate(5, sim_rw()), main="5 RW")
```



1.4.4 MA

Example 1.4.6 (First order moving average, MA(1))**.** Series with term defined[7] as

$$X_t = \theta \varepsilon_{t-1} + \varepsilon_t, \quad t = 0, \pm 1, \ldots$$

with $\{\varepsilon_t\} \sim \mathrm{WN}\left(0, \sigma^2\right)$ and $\theta \in \mathbb{R}$.
The process is *stationary* since

$$\begin{aligned}
\mathbb{E}\left[X_t\right] &= \mathbb{E}\left[\varepsilon_t + \theta\varepsilon_{t-1}\right] = \mathbb{E}\left[\varepsilon_t\right] + \theta\,\mathbb{E}\left[\varepsilon_{t-1}\right] = 0 \\
\mathrm{Var}\left[X_t\right] &= \mathbb{E}\left[X_t^2\right] = \mathbb{E}\left[(\varepsilon_t + \theta\varepsilon_{t-1})^2\right] = \mathbb{E}\left[\varepsilon_t^2 + \theta^2\varepsilon_{t-1}^2 + 2\varepsilon_t\varepsilon_{t-1}\right] \\
&= \sigma^2 + \theta^2\sigma^2 + 0 = \sigma^2(1+\theta^2) < \infty \\
\mathrm{Cov}\left(X_t, X_{t+h}\right) &= \begin{cases} \sigma^2(1+\theta^2) & \text{if } h = 0 \\ \theta\sigma^2 & \text{if } h = \pm 1 \\ 0 & \text{if } |h| > 1 \end{cases}
\end{aligned}$$

For covariance, the first case is the already developed variance, the second and the third can be verified, eg for $h = 1$ and $h = 2$

$$\begin{aligned}
\mathrm{Cov}\left(X_t, X_{t+1}\right) &= \mathbb{E}\left[X_t X_{t+1}\right] - \mathbb{E}\left[X_t\right]\mathbb{E}\left[X_{t+1}\right] = \mathbb{E}\left[(\varepsilon_t + \theta\varepsilon_{t-1})(\varepsilon_{t+1} + \theta\varepsilon_t)\right] \\
&= \mathbb{E}\left[\varepsilon_t\varepsilon_{t+1} + \theta\varepsilon_t^2 + \theta\varepsilon_{t-1}\varepsilon_{t+1} + \theta^2\varepsilon_{t-1}\varepsilon_t\right] = 0 + \theta\sigma^2 + 0 + 0 = \theta\sigma^2 \\
\mathrm{Cov}\left(X_t, X_{t+2}\right) &= \ldots = \mathbb{E}\left[(\varepsilon_t + \theta\varepsilon_{t-1})(\varepsilon_{t+2} + \theta\varepsilon_{t+1})\right] \\
&= \mathbb{E}\left[\varepsilon_t\varepsilon_{t+2} + \theta\varepsilon_t\varepsilon_{t+1} + \theta\varepsilon_{t-1}\varepsilon_{t+2} + \theta^2\varepsilon_{t-1}\varepsilon_{t+1}\right] = 0 + 0 + 0 + 0
\end{aligned}$$

The autocorrelation function of $\{X_t\}$, considering $h = 0, \pm 1, \pm 2, \ldots$

$$\rho(h) = \mathrm{Corr}\left(X_t, X_{t+h}\right) = \begin{cases} 1 & \text{if } h = 0 \\ \dfrac{\theta}{(1+\theta^2)} & \text{if } h = \pm 1 \\ 0 & \text{if } |h| > 1 \end{cases}$$

Thus it's a vertical stick like function, where in -1 and $+1$ sign is the same as θ, is 1 in 0 and 0 elsewhere. So MA(1) *autocorrelation become 0 after first lag*/a

[7] La prof la definisce forzando un $-\theta$; qua per ora tengo la stessa notazione di ITSF che mi pare più semplice

single time period (the random variable are not anymore correlated). If $\theta > 0$ $then X_t$ and X_{t-1} are positively autocorrelated, otherwise negatively correlated. For calculating theoretical ACF R becomes handy, eg for $\theta = 0.9$ $\rho_1 = 0.9/(1 + 0.9^2) = 0.49$ as follows

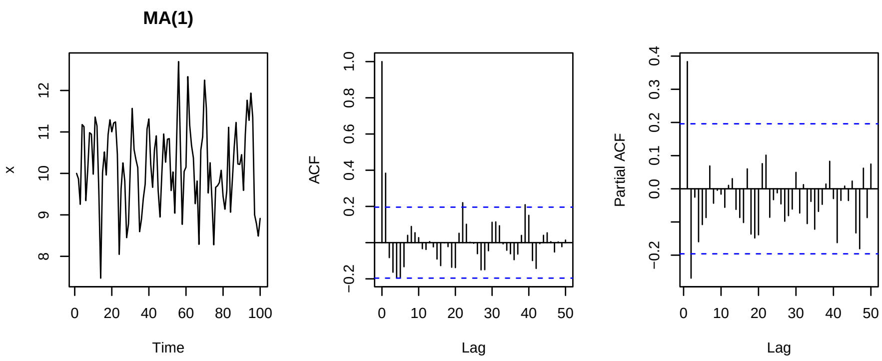
```
# calculate theoretical ACF for MA(1) model with theta=0.9
ARMAacf(ma=0.9, lag.max=10)

##      0      1      2      3      4      5      6      7      8      9     10
## 1.0000 0.4972 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000
```

Example 1.4.7 (General MA(1))**.** We can add a generic $\omega \in \mathbb{R}$ (above fixed to 0) at beginning

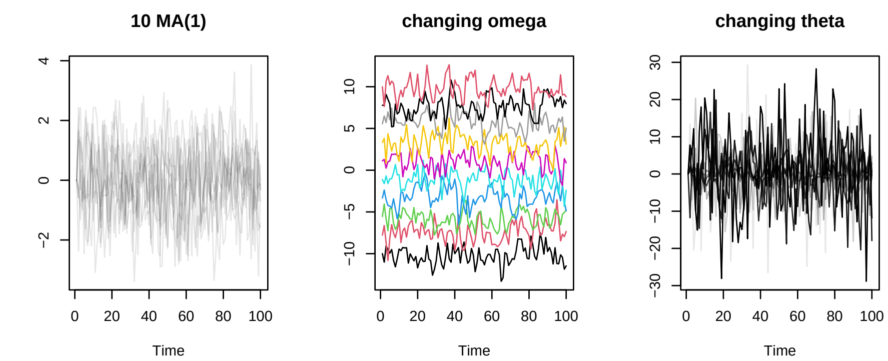
$$X_t = \omega + \theta\varepsilon_{t-1} + \varepsilon_t$$

One have $\mathbb{E}[X_t] = \omega$ and then for variance/covariance ....
Generally speaking an MA model expresses the current value of a time series as a linear function of current and past random shocks (error terms) with finite lag length. The moving-average model relies solely on the dependency structure of the error terms not on values of the process (differently from AR)

```
# function to generate data
sim_ma1

## function (n = 100, omega = 0, theta = 0.5, sigma2 = 1)
## {
##     e <- sqrt(sigma2) * rnorm(n)
##     x <- omega
##     for (t in seq(1, n - 1)) {
##         x[t + 1] <- omega + theta * e[t] + e[t + 1]
##     }
##     x
## }
## <bytecode: 0x59bc35098a80>
## <environment: namespace:lbts>

set.seed(1)
x <- sim_ma1(omega=10)
ts_plot(x, main="MA(1)")
```




```
par(mfrow=c(1,3))
mts_plot(replicate(10, sim_ma1()), main="10 MA(1)") # 10 MA(1) starting from 0
mts_plot(lapply(seq(-10, 10, length=10), function(o) sim_ma1(omega=o)),
         main = "changing omega",
         col = 1:10)
mts_plot(lapply(seq(-10, 10, length=10), function(t) sim_ma1(theta=t)),
         main = "changing theta",
         # chiari theta negativo, scuri positivo
         col = lbmisc::col2hex("black", seq(0.1, 1, length=10)))
```



Example 1.4.8 (Generic MA(q)). Defined as linear combination of q past random shock plus lift (ω) and a current error as follows:

$$X_t = \omega + \varepsilon_t + \theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2} + \ldots + \theta_q\varepsilon_{t-q}$$

with $\{\varepsilon_t\} \sim \mathrm{WN}\left(0, \sigma^2\right)$ has **ACF that becomes zero** after the q-th lag

Proposition 1.4.1. *If one has a stationary process and*

- *the autocorrelation function becomes 0 after the first lag then one can conclude that the process is an MA(1).*
- *the autocorrelation function becomes 0 after q lag then one can conclude that the process is an MA(q).*

1.4.5 AR

Example 1.4.9 (First order autoregression, AR(1)). Assume X_t is a stationary series satisfying

$$X_t = \phi X_{t-1} + \varepsilon_t, \quad t = 0, \pm 1, \dots$$

with

- $\varepsilon_t \sim \mathrm{WN}(0, \sigma)$,
- $|\phi| < 1$ (we will see that *only if* $|\phi| < 1$ then X_t is *stationary*)
- ε_t uncorrelated with X_s, for $t > s$

Taking expectation to both

$$\begin{aligned} \mathbb{E}[X_t] &= \phi\, \mathbb{E}[X_{t-1}] + \mathbb{E}[\varepsilon_t] \\ \mathbb{E}[X_t] &= \phi\, \mathbb{E}[X_{t-1}] \end{aligned}$$

Thus by stationarity $\mathbb{E}[X_t] = \mathbb{E}[X_{t-1}]$ and we conclude that $\mathbb{E}[X_t] = 0$.
To obtain autocorrelation we multiply each side by X_{t-h} and take expectation obtaining a relationship between autocovariances, and then develop using covariance properties:

$$\begin{aligned} X_t X_{t-h} &= \phi X_{t-1} X_{t-h} + \varepsilon_t X_{t-h} \\ \mathbb{E}[X_t X_{t-h}] &= \mathbb{E}[\phi X_{t-1} X_{t-h} + \varepsilon_t X_{t-h}] \\ \mathrm{Cov}(X_t, X_{t-h}) &= \phi \cdot \mathbb{E}[\phi X_{t-1} X_{t-h}] + \underbrace{\mathbb{E}[\varepsilon_t X_{t-h}]}_{\mathrm{Cov}(\varepsilon_t, X_{t-h}) = 0} \\ &= \phi \cdot \mathrm{Cov}(X_{t-1}, X_{t-h}) \\ \gamma(h) &= \phi \cdot \gamma(h-1) \\ &= \phi \cdot \phi \cdot \gamma(h-2) = \dots \\ &= \phi^h \cdot \underbrace{\gamma(0)}_{\mathrm{Var}[X_t]} \end{aligned}$$

thus the autocorrelation

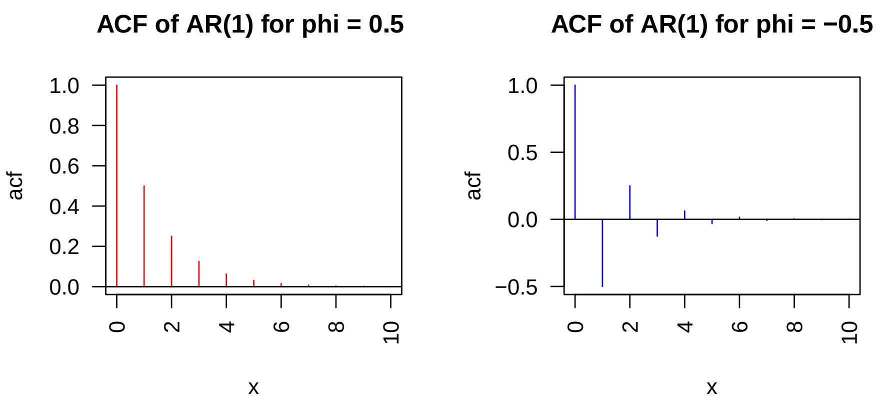
$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \phi^h$$

and being in general that covariance is symmetric $\gamma(h) = \gamma(-h)$ we can write/it must be that the autocorrelation

$$\rho(h) = \phi^{|h|}, \quad h = 0, \pm 1, \pm 2, \dots$$

So for $|\phi| < 1$ shape of autocorrelation with lag h is a stick function which is decreasing if $\phi > 0$ and is oscillating around 0 if $\phi < 0$. In both cases never goes to 0, differently from MA(1). For AR the most common case is having $0 < \phi < 1$ (positive decreasing autocorrelation).

```
ar1_rho <- function(phi=0.5, h=0:10) phi^(abs(h))

par(mfrow=c(1,2))
stick_plot(ar1_rho(), main= "ACF of AR(1) for phi = 0.5 ", col = "red",
           xlab="Time", ylab="acf", las=2)
stick_plot(ar1_rho(phi = -0.5), main= "ACF of AR(1) for phi = -0.5", col = "blue",
           xlab="Time", ylab="acf", las=2)
```




```
## for theoretical acf one can use ARMAacf:
# ACF of AR(1) with parameter phi = 0.5
ARMAacf(ar=0.5, lag.max=10)

##         0         1         2         3         4         5         6         7         8
## 1.0000000 0.5000000 0.2500000 0.1250000 0.0625000 0.0312500 0.0156250 0.0078125 0.0039062
##         9        10
## 0.0019531 0.0009766

# ACF of AR(1) with parameter phi = -0.5
ARMAacf(ar=-0.5, lag.max=10)

##          0          1          2          3          4          5          6          7
##  1.0000000 -0.5000000  0.2500000 -0.1250000  0.0625000 -0.0312500  0.0156250 -0.0078125
##          8          9         10
##  0.0039062 -0.0019531  0.0009766
```

Regarding variance:

$$\begin{aligned}
\mathrm{Var}\left[X_t\right] &= \mathrm{Cov}\left(X_t, X_t\right) = \mathrm{Cov}\left(\phi X_{t-1} + \varepsilon_t, \phi X_{t-1} + \varepsilon_t\right) \\
&= \phi\,\mathrm{Cov}\left(X_{t-1}, \phi X_{t-1} + \varepsilon_t\right) + \mathrm{Cov}\left(\varepsilon_t, \phi X_{t-1} + \varepsilon_t\right) \\
&= \phi\left[\phi\,\underbrace{\mathrm{Cov}\left(X_{t-1}, X_{t-1}\right)}_{\mathrm{Var}[X_{t-1}]} + \underbrace{\mathrm{Cov}\left(X_{t-1}, Z\right)}_{=0}\right] + \phi\,\underbrace{\mathrm{Cov}\left(\varepsilon_t, X_{t-1}\right)}_{=0} + \underbrace{\mathrm{Cov}\left(\varepsilon_t, \varepsilon_t\right)}_{=\mathrm{Var}[\varepsilon_t]=\sigma^2} \\
&= \phi^2\,\mathrm{Var}\left[X_{t-1}\right] + \sigma^2
\end{aligned}$$

Being stationary by assumption $\mathrm{Var}\left[X_{t-1}\right] = \mathrm{Var}\left[X_t\right]$ thus

$$\begin{aligned}\mathrm{Var}\left[X_t\right] &= \phi^2 \,\mathrm{Var}\left[X_t\right] + \sigma^2 \\ \mathrm{Var}\left[X_t\right]\left(1-\phi^2\right) &= \sigma^2\end{aligned}$$

and thus

$$\gamma_0 = \mathrm{Var}\left[X_t\right] = \frac{\sigma^2}{1-\phi^2}$$

so we need $\phi \neq 1$ for γ_0 to exist

Example 1.4.10 (General AR(1))**.** Adding a generic $\omega \in \mathbb{R}$

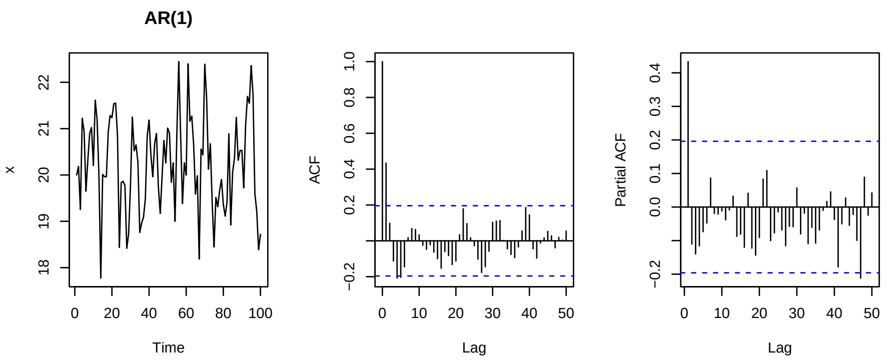
$$X_t = \omega + \phi X_{t-1} + \varepsilon_t$$

Thus here model specifies current value of the process as dependent linearly on past values plus error terms.
We have that and for (hypothysed) stationarity $\mathbb{E}\left[X_t\right] = \mathbb{E}\left[X_{t-1}\right] = \mu_t = \mu$ so

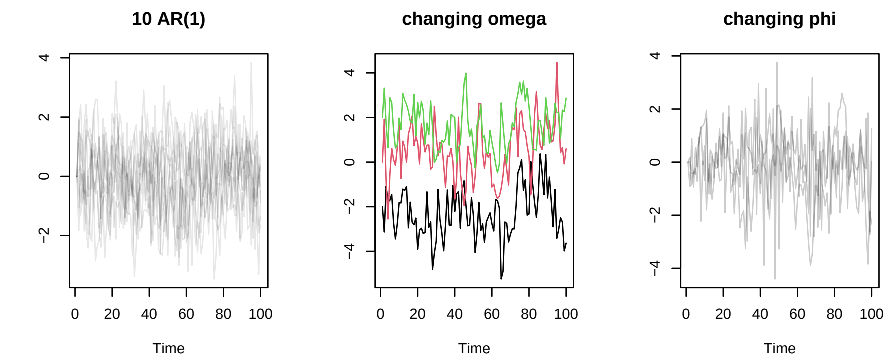
$$\begin{aligned}\mathbb{E}\left[X_t\right] &= \omega + \phi\,\mathbb{E}\left[X_{t-1}\right] \\ \mu &= \omega + \phi\mu \\ \mu &= \frac{\omega}{1-\phi}\end{aligned}$$

with $\phi \neq 1$. In the following we assume $\omega = 0$ thus $\mathbb{E}\left[X_t\right] = 0$.

```r
sim_ar1

## function (n = 100, omega = 0, phi = 0.5, sigma2 = 1)
## {
##     e <- sqrt(sigma2) * rnorm(n)
##     x <- omega/(1 - phi)
##     for (t in seq(1, n - 1)) {
##         x[t + 1] <- omega + phi * x[t] + e[t + 1]
##     }
##     x
## }
## <bytecode: 0x59bc383eb4f8>
## <environment: namespace:lbts>

set.seed(1)
x <- sim_ar1(omega=10)
ts_plot(x, main="AR(1)")
```

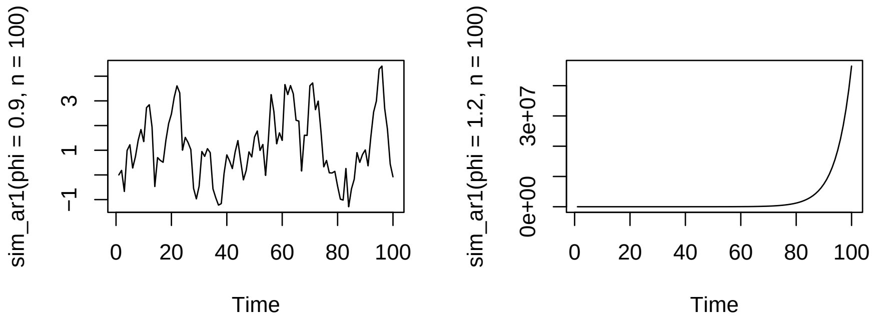



```
par(mfrow=c(1,3))
mts_plot(replicate(10, sim_ar1()), main="10 AR(1)") # 10 MA(1) starting from 0
mts_plot(lapply(c(-1,0,1), function(o) sim_ar1(omega=o)),
       main = "changing omega",
       col = 1:3)
mts_plot(lapply(c(-0.8,0,0.8), function(p) sim_ar1(phi=p)),
       main = "changing phi",
       col = lbmisc::col2hex("black", seq(0.2, 0.8, 3)))
```



Finally let's see for $\omega = 0$ the impact of changing $\theta = 0.99$ to 1 (the series is no more stationary)

```
set.seed(1)
par(mfrow=c(1,2))
ts.plot(sim_ar1(phi=0.90, n=100))
ts.plot(sim_ar1(phi=1.2, n=100))
```



Example 1.4.11 (Generic AR(p) process)**.** Defined as linear combination of p past values plus lift (ω) and an error as follows:

$$X_t = \omega + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \ldots + \phi_p X_{t-p} + \varepsilon_t$$

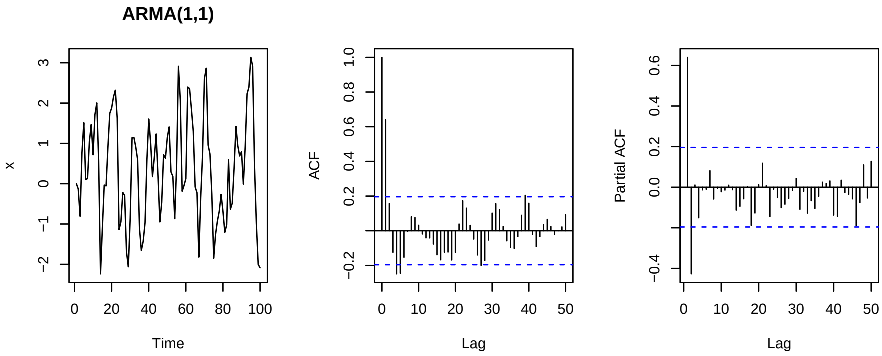
with $\{\varepsilon_t\} \sim \text{WN}\left(0, \sigma^2\right)$ has an ACF that never goes to zero. So differently from MA, we cannot identify the order of the autoregressive process looking at its ACF

Example 1.4.12 (ARMA(1,1) process)**.** Defined as a fusion of AR(1) and MA(1) a process such that

$$X_t = \omega + \phi X_{t-1} + \theta \varepsilon_{t-1} \varepsilon_t$$

with $\{\varepsilon_t\} \sim \text{WN}\left(0, \sigma^2\right)$ $\theta \in R$ and the process is stationary for $|\phi| < 1$

```
sim_arma11

## function (n = 100, omega = 0, theta = 0.5, phi = 0.5, sigma2 = 1)
## {
##     e <- sqrt(sigma2) * rnorm(n)
##     x <- omega/(1 - phi)
##     for (t in seq(1, n - 1)) {
##         x[t + 1] = omega + phi * x[t] + theta * e[t] + e[t +
##             1]
##     }
##     x
## }
## <bytecode: 0x59bc3597a8c0>
## <environment: namespace:lbts>

set.seed(1)
x <- sim_arma11()
ts_plot(x, main="ARMA(1,1)")
```



1.4.6 Other R utilities

```
# --------------
# SIMULATE DATA
# --------------
## all the process below with mean 1
mu <- 1
set.seed(123)

# MA(1) processes with theta 0.9 and e(t) ~ N(0,1)
ma1.sim <- mu + arima.sim(model=list(ma=0.9), n=250)
# MA(1) process with theta 0.9 and e(t) ~ N(0,(0.1)^2)
ma1.sim2 <- mu + arima.sim(model=list(ma=0.9), n=250, innov=rnorm(n=250, mean=0, sd=0.1))

# AR(1) with parameter phi=0.9
ar1.sim <- mu + arima.sim(model=list(ar=0.9), n=250)


## --------------------------
## theoretical ACF (examples)
## --------------------------

# ACF for AR(1) model: what happens if phi increase in absolute value
ARMAacf(ar=0.75, lag.max=5)

##      0      1      2      3      4      5
## 1.0000 0.7500 0.5625 0.4219 0.3164 0.2373

ARMAacf(ar=0.99, ma=0, lag.max=5) # corr increase

##      0      1      2      3      4      5
## 1.0000 0.9900 0.9801 0.9703 0.9606 0.9510

ARMAacf(ar=-0.75, lag.max=5)

##       0       1       2       3       4       5
##  1.0000 -0.7500  0.5625 -0.4219  0.3164 -0.2373
```

```
ARMAacf(ar=-0.99, ma=0, lag.max=5) # corr increase in absolute value

##       0       1       2       3       4       5
##  1.0000 -0.9900  0.9801 -0.9703  0.9606 -0.9510

# ACF for AR(2) model with phi_1 = 0.75 phi_2 = 0.2
ARMAacf(ar=c(0.75,.2), lag.max=10)

##      0      1      2      3      4      5      6      7      8      9     10
## 1.0000 0.9375 0.9031 0.8648 0.8293 0.7949 0.7620 0.7305 0.7003 0.6713 0.6435
```

1.5 Sample autocorrelation function

Remark 8. Previously we've seen autocorrelation function for theoretical time series models.
In practical problems we do not start with a model but with observed data $x_1, x_2, \dots, x_n$. To assess the degree of dependence in the data and to select a model for the data that reflects this, one of the important tools we use is the **sample autocorrelation function** (sample ACF) of the data.
If we we believe that the data are realized values of a stationary time series $\{X_t\}$ then the sample ACF will provide us with an estimate of the ACF of $\{X_t\}$. This estimate may suggest which of the many possible stationary time series models is a suitable candidate for representing the dependence in the data.
For example, a sample ACF that is close to zero for all nonzero lags suggests that an appropriate model for the data might be iid noise.
Let's see sample analogues for autocovariance and autocorrelation

Definition 1.5.1. If $x_1, \dots, x_n$ are observation of a time series, the sample mean is defined as

$$\overline{x} = \frac{\sum_{t=1}^{n} x_t}{n}$$

Definition 1.5.2 (Sample autocovariance function)**.** Defined as for h such that $-n < h < n$

$$\widehat{\gamma}(h) = \frac{\sum_{t=1}^{n-|h|}(x_{t+|x|} - \overline{x})(x_t - \overline{x})}{n}$$

Remark 9. for $h \geq 0$, $\widehat{\gamma}(h)$ is approximately equal to the sample covariance of the $n-h$ pairs of observations $(x_1, x_{1+h}), (x_2, x_{2+h}), \dots (x_{n-h}, x_n)$, with only difference in the denominator (n instead of $n-h$, to ensure that sample covariance matrix $\widehat{\Gamma_n} = [\widehat{\gamma}(i-j)]_{i,j=1}^{n}$ is nonnegative definite) and the subtraction of the overall mean $\overline{x}$ from each factor of the summand

Definition 1.5.3 (Sample autocorrelation function)**.** Again defined for h such that $-n < h < n$ as

$$\widehat{\rho}(h) = \frac{\widehat{\gamma}(h)}{\widehat{\gamma}(0)}$$

Remark 10. Like sample covariance matrix, sample correlation matrix $\widehat{R_n} = [\widehat{\rho}(i-j)]_{i,j=1}^n$ is nonnegativie definite and each of its diagonal elements is equal to 1

Important remark 4. The sample autocovariance and autocorrelation functions can be computed for any data set and are not restricted to observations from a stationary time series.
For data

- containing a trend, $|\widehat{\rho}(h)|$ will exhibit slow decay as h increases
- with a substantial deterministic periodic component, $|\widehat{\rho}(h)|$ will exhibit similar behavior with the same periodicity.

Thus $\widehat{\rho}$ can be useful as indicator of nonstationarity

Chapter 2

Linear processes

2.1 Linear process

Remark 11. We now see a family of processess which encompass AR and MA under some conditions.
Relevance of this process is related to Wold decomposition theorem and Wold representation

Definition 2.1.1 (Linear process (MA(∞)))**.** A process defined like infinite linear combination of variables that form a white noise process:

$$X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j} \tag{2.1}$$

where

- $\varepsilon_t \sim \mathrm{WN}\left(0, \sigma^2\right)$
- $\psi_0 = 1$ and $\sum_{j=0}^{\infty} \psi_j^2 < \infty$

Linear processes are also referred to as MA(∞) processes

Proposition 2.1.1 (Properties)**.** *A linear process has*

$$\begin{aligned}
\mathbb{E}\left[X_t\right] &= 0 \\
\gamma_0 = \operatorname{Var}\left[X_t\right] &= \sigma^2 \sum_{j=0}^{\infty} \psi_j^2 \quad < \infty \textit{(by def)} \\
\gamma_k = \operatorname{Cov}\left(X_t, X_{t+k}\right) &= \sigma^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+|k|}
\end{aligned}$$

Thus linear process are stationary process.

Proof. Respectively for mean

$$\mathbb{E}\left[X_t\right] = \mathbb{E}\left[\sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}\right] = \sum_{j=0}^{\infty} \psi_j \, \mathbb{E}\left[\varepsilon_{t-j}\right] = 0$$

where at the last, we needed the sum of squared coefficients to be convergent.
For variance:

$$
\begin{aligned}
\gamma_0 = \operatorname{Var}[X_t] = \mathbb{E}[X_t X_t] = \mathbb{E}\left[(\sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j})^2\right] \\
&= \mathbb{E}[(\varepsilon_t + \psi_1\varepsilon_{t-1} + \psi_2\varepsilon_{t-2} + \ldots)(\varepsilon_t + \psi_1\varepsilon_{t-1} + \psi_2\varepsilon_{t-2} + \ldots)] \\
&= \mathbb{E}\left[\varepsilon_t^2 + \varepsilon_t(\psi_1\varepsilon_{t-1} + \psi_2\varepsilon_{t-2} + \ldots) + \psi_1\varepsilon_{t-1}\varepsilon_t + \psi_1^2\varepsilon_{t-1}^2 + \psi_1\varepsilon_{t-1}(\psi_1\varepsilon_{t-2} + \ldots) + \ldots\right] \\
&= \sigma^2 + 0 + 0 + \psi_1^2\sigma^2 + 0 + 0 + \psi_2^2\sigma^2 + \ldots \\
&= \sigma^2 \sum_{j=0}^{\infty} \psi_j^2
\end{aligned}
$$

which is $< \infty$ by assumption of definition.
For covariance

$$
\begin{aligned}
\gamma_1 &= \mathbb{E}[X_t X_{t+1}] \\
&= \mathbb{E}[(\varepsilon_t + \psi_1\varepsilon_{t-1} + \psi_2\varepsilon_{t-2} + \ldots)(\varepsilon_{t+1} + \psi_1\varepsilon_t + \psi_2\varepsilon_{t-1} + \ldots)] \\
&= \mathbb{E}\left[\varepsilon_t\varepsilon_{t+1} + \psi_1\varepsilon_t^2 + \psi_2\varepsilon_t\varepsilon_{t-1} + \ldots + \psi_1\varepsilon_{t-1}\varepsilon_{t+1} + \psi_1^2\varepsilon_{t-1}\varepsilon_t + \psi_1\psi_2\varepsilon_{t-1}^2 + \ldots + \psi_2\varepsilon_{t-2}\varepsilon_{t+1} + \psi_2\psi_1\right. \\
&= \mathbb{E}\left[\psi_1\varepsilon_t^2 + \psi_1\psi_2\varepsilon_{t-1}^2\right] + \psi_2\psi_3\varepsilon_{t-1} + \ldots \\
&= \psi_1\sigma^2 + \psi_1\psi_2\sigma^2 + \psi_2\psi_3\sigma^2 + \ldots \\
&= \sigma^2 \sum_{j=0}^{\infty} \psi_j\psi_{j+1}
\end{aligned}
$$

and the last due to $\psi_0 = 1$.
For the second level

$$
\gamma_2 = \mathbb{E}[X_t X_{t+2}] = \ldots = \sigma^2(\psi_0\psi_2 + \psi_1\psi_3 + \ldots) = \sigma^2 \sum_{j=0}^{\infty} \psi_j\psi_{j+2}
$$

□

Theorem 2.1.2 (Wold decomposition theorem)**.** *Any stationary stochastic process* $\{Z_t : t \in \mathbb{Z}\}$ *can be written/decomposed as the sum of two processes:*

$$
Z_t = X_t + D_t \quad t \in \mathbb{Z}
$$

where:

- $\{X_t : t \in \mathbb{Z}\}$ *is a* linear process
- $\{D_t : t \in \mathbb{Z}\}$ *is a* deterministic process*, defined as* $D_t = a + b \cdot t$*, with* $a, b \in \mathbb{R}$ *known (basically it's a process with zero variance at each time)*

Example 2.1.1 (Wold representation of MA(1) with zero mean)**.** The process is

$$
X_t = \varepsilon_t + \theta\varepsilon_{t-1}
$$

with $\varepsilon_t \sim \text{WN}\left(0, \sigma^2\right)$. The linear Wold representation requires that a process is writable as

$$X_t = \psi_0 \varepsilon_t + \psi_1 \varepsilon_{t-1} + \psi_2 \varepsilon_{t-2} + \dots$$

Thus for an MA(1) we can just equate above and the wold representation of a MA(1) is

$$\begin{aligned}
\psi_0 &= 1 \\
\psi_1 &= \theta \\
\psi_k &= 0, \quad \forall k > 1
\end{aligned}$$

This above is the *Wold reprentation of a MA(1)*
By using general results regarding linear processes one can retrieve what already found previously, for example Cov (X_t, X_{t+1})

$$\begin{aligned}
\gamma_1 &= \sigma^2 \left(\sum_{j=0}^{\infty} \psi_j \cdot \psi_{j+1} \right) \\
&= \sigma^2 (\psi_0 \psi_1 + \psi_1 \psi_2 + \psi_2 \psi_3 + \dots) \\
&= \sigma^2 (\theta + \theta \cdot 0 + 0 + 0 + \dots) \\
&= \sigma^2 \theta
\end{aligned}$$

while for variance

$$\begin{aligned}
\gamma_0 = \sigma^2 \left(\sum_{j=0}^{\infty} \psi_j \cdot \psi_{j+1} \right) &= \sigma^2 (1 + \psi_1^2 + \psi_2^2 + \dots) \\
= \sigma^2 (1 + \theta^2 + 0 + \dots) &= \sigma^2 (1 + \theta^2)
\end{aligned}$$

Example 2.1.2 (Wold representation of AR(1) with zero mean)**.** For MAs is easy, it will be more useful for AR processes. For AR(1) we hypothesized it's a stationary process; being able to write it in its Wold will give us the condition under which it's stationary.
The model is

$$X_t = \phi X_{t-1} + \varepsilon_t, \quad \varepsilon \sim \text{WN}\left(0, \sigma^2\right)$$

To put it in Wold representation we have to rewrite it as linear combination of white noise (and not as function of past terms of the series itself):

$$X_t = \psi_0 \varepsilon_t + \psi_1 \varepsilon_{t-1} + \psi_2 \varepsilon_{t-2} + \dots$$

To do so we can at least in first instance use recursion, by having

$$X_{t-1} = \phi X_{t-2} + \varepsilon_{t-1}$$

and by replacement we get

$$X_t = \phi(\phi X_{t-2} + \varepsilon_{t-1}) + \varepsilon_t = \phi^2 X_{t-2} + \phi \varepsilon_{t-1} + \varepsilon_t$$

Going on with recursion

$$X_{t-2} = \phi X_{t-3} + \varepsilon_{t-2}$$

and thus again by substitution in X_t formula

$$X_t = \phi^2(\phi X_{t-3} + \varepsilon_{t-2}) + \phi\varepsilon_{t-1} + \varepsilon_t = \phi^3 X_{t-3} + \phi^2 \varepsilon_{t-2} \phi\varepsilon_{t-1} + \varepsilon_t$$

And going like that, going ahead in general we obtain:

$$X_t = \phi^n X_{t-n} + \sum_{j=0}^{n-1} \phi^j \varepsilon_{t-j}$$

This above can be the representation of a linear process (and thus being a stationary process) if and only if $|\phi| < 1$ (what we've asked for in the AR definition). Infact if that is the case, for $n \to \infty$ we have that $\phi^n \to 0$ and AR(1) becomes

$$X_t = \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}$$

Obtaining its Wold representation is now easy by having

$$\psi_j = \phi^j$$

So under these condition even AR(1) is a linear process and we can find what already found via linear process properties. Eg for autocorrelatio γ_k

$$\begin{aligned}
\gamma_k &= \sigma^2 \left(\sum_{j=0}^{\infty} \psi_j \psi_{j+|k|} \right) = \sigma^2 \sum_{j=0}^{\infty} \phi^j \phi^{j+|k|} = \sigma^2 \sum_{j=0}^{\infty} \phi^j \phi^j \phi^{|k|} \\
&= \sigma^2 \phi^{|k|} \cdot \sum_{j=0}^{\infty} \phi^{2j} = \sigma^2 \phi^{|k|} \cdot \sum_{j=0}^{\infty} \left(\phi^2\right)^j \overset{(1)}{=} \sigma^2 \frac{\phi^{|k|}}{1-\phi^2} \\
&= \frac{\sigma^2}{1-\phi^2} \cdot \phi^{|k|}
\end{aligned}$$

where in (1) $\sum_{j=0}^{\infty} \left(\phi^2\right)^j \to \frac{1}{1-\phi^2}$ (geometric series) if $|\phi| < 1$

2.2 Polynomials with lag operator

Remark 12. There's a way to *write processes in a bit more compact form*, it involves the use of backshift operator (indicated by B), sometimes called also lag operator (L)

Definition 2.2.1 (Lag operator)**.** Applied to a random variable of random process it returns the previous one

$$L \cdot X_t = X_{t-1}$$

Proposition 2.2.1. *In notation, it works similarly to algebra*

$$L^k X_t = X_{t-k}$$
$$L^{-k} X_t = X_{t+k}$$

Definition 2.2.2 (Difference operator)**.** Defined using the lag as $(1 - L)$ that is

$$(1 - L)X_t = X_t - X_{t-1}$$

Remark 13. It's used to remove trend and autocorrelation, but we're not limited to use it for a single period; eg if seasonality is available, value from the same month could be correlated and we could be interested in looking at seasonal difference operator

Definition 2.2.3 (Seasonal difference operator)**.** Defined as $(1 - L^s)$ with s length of the season (tipically $s = 12$ for monthly data or $s = 4$ for quarterly data).

$$(1 - L^{12})X_t = X_t - X_{t-12}$$

Proposition 2.2.2 (Second order difference operator)**.** *For further differentiation difference operator works like algebra; to do a double differentiation*[1] *we take the first difference and then difference again*

$$\begin{aligned}(1 - L)^2 X_t &= (1 - L)\underbrace{(1 - L)X_t}_{\text{first}} \\ &= (1 - 2L + L^2)X_t \\ &= X_t - 2X_{t-1} + X_{t-2}\end{aligned}$$

Pay attention that $(1 - L)^2 \neq (1 - L^2)$

2.3 Polynomials in the lag operator

Remark 14. This simplify a lot computations

Important remark 5. A linear process can be rewritten using the lag operator as

$$\begin{aligned}X_t &= \varepsilon_t + \psi_1\varepsilon_{t-1} + \psi_2\varepsilon_{t-2} + \dots \\ &= L^0\varepsilon_t + \psi_1 L\varepsilon_t + \psi_2 L^2\varepsilon_t + \dots \\ &= (1 + \psi_1 L + \psi_2 L^2 + \dots) \cdot \varepsilon_t \\ &= \Psi(L) \cdot \varepsilon_t\end{aligned}$$

where

$$\Psi(L) = 1 + \psi_1 L + \psi_2 L^2 + \dots$$

with $\Psi(L)$ a polynomial (order ∞) in the lag operator, with coefficients $1, \psi_1, \psi_2, \dots$

[1] Similarly to second order derivative we obtain a constant if say we start from a parabola; second order differences allow us to make stationary a series which is non stationary because of the presence of a polynomial trend typical of order 2. It's not that frequent to apply a second order difference

Example 2.3.1 (MA(1) rewritten)**.** we have

$$X_t = \varepsilon_t + \theta\varepsilon_{t-1} = \varepsilon_t + \theta L\varepsilon_t = (1+\theta L)\varepsilon_t$$

Example 2.3.2 (AR(1) rewritten)**.** We have

$$X_t = \phi X_{t-1} + \varepsilon_t X_t - \phi L X_t = \varepsilon_t (1-\phi L)X_t = \varepsilon_t$$

2.4 ARMA process

Example 2.4.1 (MA(q))**.** We have

$$\begin{aligned} X_t &= \varepsilon_t + \theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2} + \dots \theta_q\varepsilon_{t-q} = (1+\theta_1 L + \theta_2 L^2 + \dots + \theta_q L^q)\varepsilon_t \\ &= \Theta_q(L)\varepsilon_t \end{aligned}$$

with

$$\Theta_q(L) = 1 + \theta_1 L + \theta_2 L^2 + \dots + \theta_q L^q$$

Example 2.4.2 (AR(p))**.**

$$\begin{aligned} &X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + \varepsilon_t \\ &(1 - \phi_1 L - \phi_2 L^2 - \dots - \phi_p L^p)X_t = \varepsilon_t \\ &\Phi_p(L)X_t = \varepsilon_t \end{aligned}$$

with

$$\Phi_p(L) = 1 - \phi_1 L - \phi_2 L^2 - \dots - \phi_p L^p$$

Remark 15. We may have both components

Example 2.4.3 (ARMA(1,1))**.** Defined as

$$(1-\phi L)X_t = (1+\theta L)\varepsilon_t$$

That developed leads to

$$X_t = \phi X_{t-1} + \varepsilon_t + \theta\varepsilon_{t-1}$$

where $\phi X_{t-1} + \varepsilon_t$ is the AR(1) part and $\varepsilon_t + \theta\varepsilon_{t-1}$ the MA(1)

Example 2.4.4 (ARMA(p,q))**.** Defined as

$$\Phi_p(L)X_t = \Theta_q(L)\varepsilon_t$$

that developed leads to

$$X_t = \underbrace{\phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p}}_{AR(p)} + \varepsilon_t + \underbrace{\theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2} + \dots \theta_q\varepsilon_{t-q}}_{MA(q)}$$

2.5 Invertibility

Definition 2.5.1. A stochastic process $\{X_t : t \in \mathbb{Z}\}$ is invertible if it can be expressed as function of its past values plus an *innovation term* $\varepsilon_t \sim \text{WN}\left(0,\sigma^2\right)$

$$X_t = g(X_{t-1}, X_{t-2}, \dots; \varepsilon_t) \quad \varepsilon_t \sim \text{WN}\left(0,\sigma^2\right)$$

2.5.1 AR(1) and MA(1)

Example 2.5.1 (Invertibility of AR(1))**.** It's clearly an invertible

$$X_t = \omega + \phi X_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \text{WN}\left(0, \sigma^2\right)$$

Example 2.5.2 (Invertibility of MA(1))**.** Less clearly, the MA:

$$X_t = \omega + \theta \varepsilon_{t-1} + \varepsilon_t, \quad X_t \sim \text{WN}\left(0, \sigma^2\right)$$

it is not direclty written as an invertible process (since does not depend on X_{t-1} etc). We could do the following by substitution

$$\begin{aligned}
\varepsilon_t &= X_t - \omega - \theta \varepsilon_{t-1} \\
\varepsilon_{t-1} &= X_{t-1} - \omega - \theta \varepsilon_{t-2}
\end{aligned}$$

having

$$X_t = \omega + \theta (X_{t-1} - \omega - \theta \varepsilon_{t-2}) + (X_t - \omega - \theta \varepsilon_{t-1})$$

and going on with substitution of $\varepsilon_{t-1}, \varepsilon_{t-2}, \ldots$

Remark 16. In general:

- AR process are *invertible by construction*/definition but not stationary unless we impose conditions on coefficients ϕ
- MA(1) are stationary by construction but are not directly invertible.

We however need a better general tool to derive invertibility and stationarity conditions for stochastic processes.
Invertibility of a stochastic process and/or stationarity of a stochastic process are related to the mathematical invertibility of the polynomial in the lag operator that is associated with the process.

Proposition 2.5.1 (Invertibility di MA(1))**.** *Let's tackle invertibility conditions for MA(1) having mean $\omega = 0$. Considering $\omega = 0$*

NB: qui adotto la sua definizione malata di MA con il segno meno del coefficiente perché non ho tempo

$$\begin{aligned}
X_t &= \varepsilon_t - \theta \varepsilon_{t-1}, \quad \varepsilon_t \sim \text{WN}\left(0, \sigma^2\right) \\
&= (1 - \theta L) \varepsilon_t
\end{aligned}$$

Here we've expressed the process a function of a polynomial in the lag operator. Now

1. *in general, adopting the Wold representation we can write the process with a generic $MA(\infty)$ representation*

   $$X_t = \Theta(L) \cdot \varepsilon_t$$

   in practice we have a polynomial in the lag operator $\Theta(L)$ that acts on ε_t so one end up with X_t being a linear combination of current and past values of ε_t

2. *but for invertibility one would need MA to be writable as*

$$\Pi(L)X_t = \varepsilon_t$$

because in that case it would be easy to move elements from left (function of past values of X_t) to the right.
This above is called $AR(\infty)$ representation of MA with $\Pi(L)$ a polynomial acting on X_t like

$$\Pi(L) = 1 + \pi_1 L + \pi_2 L^2 + \dots$$

If exists the inverse $\Theta^{-1}(L)$, let's say $(1-\theta L)^{-1}$ then we could use it as $\Pi(L)$ by pre-multiplying both members of process definition, eg

$$(1-\theta L)^{-1} X_t = \varepsilon_t$$

Who can $(1-\theta L)^{-1}$ be? This is like saying

$$\frac{1}{(1-\theta L)} X_t = \varepsilon_t$$

Basically, similarly to a geometric series, if $|\theta| < 1$ we have that $\frac{1}{1-\theta L}$ is the limit to which the geometric series of reason θL converges, for $n \to \infty$:

$$\lim_{n\to\infty} \sum_{j=0}^{n} \theta^j L^j = \sum_{j=0}^{\infty} \theta^j L^j = \frac{1}{1-\theta L}, \quad \textit{with } |\theta| < 1$$

For this reason if $|\theta| < 1$ MA(1) is invertible because we can write

$$\frac{1}{1-\theta L} X_t = \varepsilon_t$$
$$\left(\sum_{j=0}^{\infty} \theta^j L^j\right) X_t = \sum_{j=0}^{\infty} \theta^j L^j X_t \varepsilon_t$$

thus we have

$$X_t + \theta X_{t-1} + \theta^2 X_{t-2} + \dots = \varepsilon_t$$
$$X_t = \varepsilon_t - \theta X_{t-1} - \theta^2 X_{t-2} - \dots$$

so above

- *we have the AR(∞) representation of a MA, it's an infinite sum but we have just* one *parameter which is θ*
- *the oefficient of the inverse $\Theta^{-1}(L)$ that is $\Pi(L)$ are simply $\pi_j = \theta^j$*
- *we have represented even MA as function of past own values and innovation term, thus it's invertible*
- *the condition of invertibility of MA(1) $|\theta| < 1$ are the same of stationarity of AR(1) process ($|\phi| < 1$)*

2.5.2 Implications of invertibility of MA(1)

Remark 17. On the practical/interpretational level invertibility is important for identification and for asymptotic forgetting of initial conditions (at a certain stage the past become irrelevant).

2.5.2.1 Identification

Focusing again on MA(1) (with zero mean, $\omega = 0 \implies \mathbb{E}[X_t] = 0$)

$$\begin{aligned} X_t &= \varepsilon_t - \theta\varepsilon_{t-1}, \quad \varepsilon_t \sim \text{WN}\left(0, \sigma^2\right) \\ &= (1 - \theta L)\varepsilon_t \end{aligned}$$

we've said it is

- always stationary
- invertible if $|\theta| < 1$ (condition of convergence of geometric series)
- the *theoretical ACF* of a MA(1) is

NB: qui è la sua notazione con MA a definta con $-\theta$

$$\rho_k = \begin{cases} 1 & \text{if } k = 0 \\ -\frac{\theta}{1+\theta^2}, & \text{if } |k| = 1 \\ 0 & \text{if } |k| > 1 \end{cases}$$

Two points regarding ACF and identification:

- the $|\theta| < 1$ condition is needed otherwise setting $\theta = x$ (eg $x = 0.4$) or setting $\theta = 1/x = 1/0.4$ would lead to the same theoretical autocorrelation (check it), and the solution would be non invertible.
  So first implication of invertibility is identification
- ACF characterize MA process: once one see the acf of a stationary process going to 0 after a certain time t it's an $MA(t)$.
  This can be a tool to identify the data generating process: basically one estimates the *empirical ACF* looking at data and if the process is stationary and the autocorrelation goes to zero after tot it's an MA(tot).
  This will tell the number of parameter of interest to estimate (eg MA(1) has only one θ and we want to estimate it)

2.5.2.2 Asymptotic forgetting/discounting the past

Back to $AR(\infty)$ representation of a $MA(1)$ process

$$X_t = \varepsilon_t - \theta X_{t-1} - \theta^2 X_{t-2} - \dots$$

if $|\theta| < 1$ then past values of X_t have smaller and smaller impact on X_t.

2.5.2.3 Implication for forecast

With the classical MA(1) representation

$$X_t = \varepsilon_t - \theta\varepsilon_{t-1}, \quad \varepsilon_t \sim \text{WN}\left(0, \sigma^2\right)$$

What one tipically observe is the X_t. If one want to make any kind of forecast one need to estimate ε_t as a function of what is observed. Via the $AR(\infty)$ representation of the MA(1) we can write the process as a function of its past observations (by def of invertibility).

2.5.3 Concluding remarks

In summary we've discussed about *properties*

- stationarity
- invertibility

According to the process we define there are *representations*:

- we would like a *stationary* process to be represented as $AR(\infty)$
- we want a Wold representation for processes which are invertible

The point is that

- $AR(\infty)$ are crucial for forecasts and identification
- Wold representation is essential for stationarity but even more for deriving the theoretical properties of a process (a linear combination of uncorrelated variable makes all the calculation easy wrt to AR(∞) representation where all the variable involved are correlated)

2.5.4 AR(p) and MA(q)

General condition (regarding parameters $\theta_1, \theta_2, \ldots$) under which MA(q) is invertible, that is writable like

$$X_t = (1 - \theta_1 L - \theta_2 L^2 - \ldots - \theta_q L)\varepsilon_t$$

equivalently we see conditions (involving $\phi_1, \ldots$) under which AR(p) is writable as

$$(1 - \phi_1 L - \phi_2 L^2 - \ldots - \phi_p L^p)X_t = \varepsilon_t$$

and *stationary*

NB: credo scrittura come Wold che implica che un processo sia stazionario

In both cases the invertibility of MA(q) and stationarity of AR(p) process depend on the invertibility of polynomials of order q and p respectively in lag operators with parameters $\theta_1, \ldots \theta_q$ and $\phi_1, \ldots, \phi_1$.
The structure of the above polynomial is the same (duality of AR and MA processes)

2.5.4.1 Invertibility of a finite polynomial

A polynomial of degree n in x (with coefficients $a_0, \ldots, a_n$)

$$P_n(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n$$

with leading term equal to one $a_0 = 1$ (convention important for us).
The characteristic equation is $P_n(x) = 0$ and has solutions/roots $z_1, z_2, \ldots, z_n$ so that we can rewrite the polynomial as the product of following terms[2]

$$P_n(x) = (z_1 - x)(z_2 - x) \cdot \ldots \cdot (z_n - x)$$

[2]eg $P_2(x) = (1 - 2x + x^2) = (x - 1)(x - 1)$ with $z_1 = 1, z_2 = 1$

Our aim is to find (if exists) the inverse of the polynomial. If we assume that roots are different from 0 $z_i \neq 0$, $i = 1, \ldots, n$ we can equivalently write it as

$$P_n(x) \propto \left(1 - \frac{x}{z_1}\right)\left(1 - \frac{x}{z_2}\right) \cdot \ldots \cdot \left(1 - \frac{x}{z_n}\right)$$

the inverse of the polynomial will be

$$(P_n(x))^{-1} \propto \left(1 - \frac{x}{z_1}\right)^{-1}\left(1 - \frac{x}{z_2}\right)^{-1} \cdot \ldots \cdot \left(1 - \frac{x}{z_n}\right)^{-1} = \prod_{i=1}^{n}\left(1 - \frac{x}{z_i}\right)^{-1}$$

All the factor are of the form $\left(1 - \frac{x}{z_i}\right)$. If we replace the variable x with the lag operator each of the these term can be considered as the limit to which the geometric series of reason $\frac{1}{z_i}$ converge provided that the modulus of the reason is <1 $\left|\frac{1}{z_i}\right| < 1$ (equivalently $|z_i| > 1$). We know by the MA(1) example that:

$$\left(1 - \frac{L}{z_i}\right)^{-1} = \left(1 - \frac{1}{z_i}L\right)^{-1} = \sum_{j=0}^{\infty}\left(\frac{1}{z_i}\right)^{j} L^j$$

if $\left|\frac{1}{z_i}\right| < 1$ eg $|z_i| > 1$.

Important remark 6. So we say that a polynomial of order n is invertible if all the roots z_i of the characteristic equation associated with the polynomial are in modulus greater than 1 (or are external to the circle of unit radius in complex terms).

This was in terms of roots of equation but we want a condition on the parameters

Example 2.5.3 (MA(1))**.**

$$X_t = (1 - \theta L)\varepsilon_t$$

The polynomial is

$$P_1(L) = \Theta_1(L) = 1 - \theta L$$

its char equation associated is

$$\begin{aligned} 1 - \theta L &= 0 \\ \theta L &= 1 \\ L &= \frac{1}{\theta} \end{aligned}$$

this last is a root/solution (and is the only one being P_1 of order 1).
The process in invertible if $P_1(L)$ is invertible and in turn $P_1(L)$ is invertible if the root $\left|\frac{1}{\theta}\right| > 1 \iff |\theta| < 1$

Example 2.5.4 (AR(1))**.** Derive the condition of stationarity of an AR(1) process (based on the roots of the AR(1) polynom), it's the same

To summarize stationarity and invertibility based on roots of characteristic equation associated with the polynomial in the lag operator, in general:

- AR(p) process are **stationary** if all the roots of $\Phi_p(L) = 0$ are > 1 in absolute value
- MA(q) process are **invertible** if all the roots of $\Theta_q(L) = 0$ are > 1 in absolute value
- mixed process: this hold as well, due to the multiplicative structure of ARMA(p, q) processes, stationarity and invertibility applies *separately* to the AR and MA polynomial.

Example 2.5.5 (ARMA(1,1))**.**

$$(1 - \phi L)X_t = (1 - \theta L)\varepsilon_t, \quad \varepsilon_t \sim \mathrm{WN}\left(0, \sigma^2\right)$$

this is stationary if $|\phi| < 1$ and invertible if $|\theta| < 1$

Roots are estimated by software and shouldnt' be greater than 1

2.5.5 Random walk a cazzo

Brief summary on AR(1) processes:

- always invertible
- stationary if $|\phi| < 1$
- if $|\phi| > 1$: brockwell davis ITSF pag 54 and 56 and remark 2 pag 57 plus problem 4.10 (credo)
- if $\phi = 1$ of AR(1) we have as special important case the random walk process, which is a *non stationary process* (given that $|\phi| < 1$ is excluded) with lot of application in analysis of stochastic trends (many series do have)

Important remark 7 (Differencing a non-stationary process)*.* The process is an AR(1) with $\phi = 1$ (so it's non stationary and its variance is infinite) that is

$$X_t = X_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathrm{WN}\left(0, \sigma^2\right)$$

suppose a random walk with t noise ($\nu = 6$ centered on 0)

$$X_t = X_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim t_6 0, 1$$

It's non stationary, but if we take $Y_t = X_t - X_{t-1}$

$$Y_t = X_t - X_{t-1} = \varepsilon_t$$

then Y_t is stationary since ε_t is a stationary process. So if we take the differences and obtain the process $\varepsilon_t \sim t_6 0, 1$ the realization will have the white noise shape. So when we take the differences of a nonstationary process (similar to take the derivative of a continuous function) what we find is a stationary process (in case of RW)

Le'ts define

$$X_t = x_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim NID(0, \sigma^2)$$

It's a non stationary process being an AR(1) with $\phi = 1$ (root on the unit circle). The differenced process is stationary:

$$(1 - L)X_t = \varepsilon_t$$

If we fix an initial condition, $X_0 = 0$ then

$$X_1 = X_0 + \varepsilon_1 = \varepsilon_1$$
$$X_2 = X_1 + \varepsilon_2 = \varepsilon_1 + \varepsilon_2 \qquad X_3 = X_2 + \varepsilon_3 = \varepsilon_1 + \varepsilon_2 + \varepsilon_3$$

and thus

$$X_t = \sum_{k=1}^{t} \varepsilon_k$$

Now with the above formulation (as sum of IID gaussian) it's easy to see $\mathbb{E}[X_t] = 0$ and $\mathrm{Var}[X_t] = t\sigma^2$ so the variance *does depend* on t (so it's a non stationary process with path produced with tends to spread out)

As exercise derive $\mathrm{Cov}(X_t, X_{t+k})$ and $\mathrm{Cov}(X_t, X_{t-k})$ (since the process is not stationary we should not expect that the two covariances are equal)

2.6 AR(2) and MA(2)

We complete the picture of basic AR and MA processes (stationarity and invertibility)

Definition 2.6.1 (zero mean AR(2))**.** Defined adding $\psi_2 X_{t-2}$ to an AR(1)

$$X_t = \phi_1 X_{t-1} + \psi_2 X_{t-2} + \varepsilon_t, \quad \varepsilon_t \sim \mathrm{WN}\left(0, \sigma^2\right)$$

rewritable as

$$(1 - \phi_1 L - \phi_2 L^2)X_t = \varepsilon_t, \quad \varepsilon_t \sim \mathrm{WN}\left(0, \sigma^2\right)$$

Remark 18. It's

- always *invertible* by construction (function of its past value and innovation term)
- *stationary* if the roots z_1, z_1 of the characteristic equation associated to the polinomial $1 - \psi_1 L - \psi_2 L^2 = 0$ are in modulus greater than one

To get this we go in a non straightforward way. We derive the covariance function

We now derive the conditions of stationarity. To derive them

- we derive γ_0 and γ_k. Yule walker recursive equations with initial conditions

$$\gamma_0 = \frac{\sigma^2(1 - \phi_2)}{(1 + \phi_2)\left[(1 - \phi_2)^2 - \phi_1^2\right]}$$
$$\gamma_1 = \frac{\phi_1 \gamma_0}{1 - \phi_2}$$

- consider that $|z_1||z_2| > 1 \implies |z_1, z_2| > 1$ and the fact that (Yule Walker equations) $\rho_1 = \phi_1\rho_0 + \phi_2\rho_1$ with $\rho_1 = \frac{\phi_1}{1-\phi_2}$

For the Wold representation solve

$$(1 - \phi_1 L - \phi_2 L^2)X_t = \varepsilon_t$$

we search for condition such that X_t can be written as

$$X_t = (1 - \phi_1 L - \phi_2 L^2)^{-1}\varepsilon_t$$

which means that the product

$$\underbrace{(1 - \phi_1 L - \phi_2 L^2)}_{\Phi_2(L)}\underbrace{(1 + \psi_1 L + \psi_2 L^2 + \ldots)}_{\Psi(L)} = 1$$

what does this product give

$$\cancel{1} + \psi_1 L + \psi_2 L^2 + \ldots - \phi_1 L - \phi_1\psi_1 L^2 - \phi_1\psi_2 L^3 - \ldots - \phi_2 L^2 - \phi_2\psi_1 L^3 - \phi_2\psi_2 L^4 + \ldots = \cancel{1}$$
$$(\psi_1 - \phi_1)L + (\psi_2 - \phi_1\psi_1 - \phi_2)L^2 + (\psi_3 - \phi_1\psi_2 - \phi_2\psi_1)L^3 + \ldots = 0$$

For all this to be 0 we have to have

$$\begin{aligned}
&\psi_1 = \phi_1 \\
&\psi_2 = \phi_1\psi_1 - \phi_2\psi_0 = \phi_1^2 - \phi_2 \\
&\psi_3 = \phi_1\psi_2 - \phi_2\psi_1 \\
&\ldots
\end{aligned}$$

Important remark 8. In general the **wold representation of an AR(2) process**

$$\psi_j = \phi_1\psi_{j-1} - \phi_2\psi_{j-2}$$

with $\psi_0 = 1$, $\psi_1 = \phi_1$. Not easy to derive conditions based on these representation

Covariance AR(2) Let's go to the covariance function of AR(2).

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$$

one way is to use wold representation but we see another way giusto per non annoiarci. The technique is the same

$$\begin{aligned}
\gamma_0 &= \mathbb{E}\left[X_t\underbrace{(\phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t)}_{X_t}\right] \\
&= \phi_1\gamma_1 + \phi_2\gamma_2 + \sigma^2
\end{aligned}$$

warning: we're assming that X_t is stationary.
Doing the same

$$\begin{aligned}\gamma_1 &= \mathbb{E}\left[X_t(\phi_1 X_t + \phi_2 X_{t-1} + \varepsilon_{t+1})\right] \\ &= \phi_1\gamma_0 + \phi_2\gamma_1 \\ \implies \gamma_1 &= \frac{\phi_1\gamma_0}{1-\phi_2}\end{aligned}$$

and

$$\begin{aligned}\gamma_2 &= \mathbb{E}\left[X_t(\phi_1 X_{t+1} + \phi_2 X_t + \varepsilon_{t+2})\right] \\ &= \phi_1\gamma_1 + \phi_2\gamma_0\end{aligned}$$

from which we develop more γ_0 by substitution

$$\gamma_0 = \frac{\phi_1^2}{1-\phi_2}\gamma_0 + \frac{\phi_1^2\phi_2}{1-\phi_2}\gamma_0 + \phi_2^2\gamma_0 + \sigma^2$$

and we do some algebra to collect for γ_0 (actually assuming $\phi_2 \neq 0$)

$$\left(1 - \frac{\psi_1^2}{1-\psi_2} - \frac{\phi_1^2\phi_2}{1-\phi_2} - \sigma_2^2\right)\gamma_0 = \sigma^2$$

$$(1-\phi_2-\phi_1^2-\phi_1^2\phi_2-(1-\phi_2)\phi_2^2)\gamma_0 = \sigma^2(1-\phi_2)$$
$$(1-\phi_2-\phi_1^2(1+\phi_2)-(1-\phi_2)\phi_2^2)\gamma_0 = \sigma^2(1-\phi_2)$$

from which doing all the calculation we end up with an expression of

$$\gamma_0 = \frac{\sigma^2(1-\phi_2)}{(1+\phi_2)((1-\phi_2)^2-\phi_1^2)}$$

Then we can go back and obtain

$$\gamma_1 = \frac{\phi_1}{1-\phi_2}\gamma_0$$

and then

$$\gamma_k = \phi_1\gamma_{k-1} + \phi_2\gamma_{k-2} \quad \text{with initial conditions } \gamma_0, \gamma_1$$

this above is the **autocovariance function of AR(2)**. We dont' want the denominator of γ_0 to be 0.

2.6.1 Stationary conditions of AR(2)

So up to now, for AR(2) process we've done

- Wold representation
- ACV assuming stationarity; Now we want to get the stationary conditions.

An AR(2) process is stationary if the roots of the characteristic equation associated with $\Phi_2(L)$ that is

$$(1 - \phi_1 L - \phi_2 L^2) = 0$$

that is the following roots

$$z_{1.2} = \frac{-\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{2\phi_2} = \begin{cases} z_1 = \frac{-\phi_1 + \sqrt{\cdots}}{\cdots} \\ z_2 = \frac{-\phi_1 - \sqrt{\cdots}}{\cdots} \end{cases}$$

are all, in modulus, greater than 1, that is

$$|z_1| > 1, |z_2| > 1(\text{thus } |z_1 z_2| > 1)$$

To have $|z_1| > 1, |z_2| > 1$ as function of ϕ-s we can setup a system

$$z_1 < -1 \vee z_1 > 1 z_2 < -1 \vee z_2 > 1$$

to find solutions by substituting $\frac{-\phi_1 \pm \sqrt{\cdots}}{\cdots}$is quite involved. But since if $|z_1| > 1, |z_2| > 1$ this implies $|z_1 z_2| > 1$, this helps because the products of the two is the type $(a+b)(a-b) = a^2 - b^2$. The product of the two will be $\frac{\phi_1^2 - \phi_1^2 - 4\phi_2}{4\phi_2^2}$ so the conditions become

$$\left| \frac{\phi_1^2 - \phi_1^2 - 4\phi_2}{4\phi_2^2} \right| > 1$$
$$|\phi_2| < 1$$

and this above, $|\phi_2| < 1$, is the first condition. But it's not enough because we need condition on ϕ_1 as well.
Remembering that we've said

$$\gamma_k = \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2}$$
$$\rho_k == \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2}$$

If $k = 1$ then

$$\rho_1 = \phi_1 \rho_0 + \phi_2 \rho_{-1}$$

assuming stationarity so $\approxeq \phi_1 \rho_0 + \phi_1 \rho_1$ so

$$\rho_1 = \phi_1 \underbrace{\rho_0}_{1} + \phi_2 \rho_1 \implies \rho_1 = \frac{\phi_1}{1 - \phi_2}$$

and since for sure

$$-1 < \rho_1 < 1$$

and $|\phi_2| < 1$ we get two additional conditions for stationarity of an AR(2)process (from the double inequality above by substituting ρ_1 developed before)

$$\frac{\phi_1}{1 - \phi_2} < 1 \implies \phi_2 + \phi_1 < 1$$

On the other hand

$$\frac{\phi_1}{1-\phi_2} > -1 \implies \phi_2 - \phi_1 < 1$$

Thus summarizing AR(2) is stationary if

$$\begin{cases} |\phi_2| < 1 \\ \phi_2 + \phi_1 < 1 \\ \phi_2 - \phi_1 < 1 \end{cases}$$

2.6.2 Stationary conditions of MA(2)

The process is

$$X_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2}, \quad \varepsilon_t \sim \mathrm{WN}\left(0, \sigma^2\right)$$

or

$$X_t = (1 - \theta_1 L \theta_2 L^2)\varepsilon_t \varepsilon_{t-2}, \quad \varepsilon_t \sim \mathrm{WN}\left(0, \sigma^2\right)$$

The stationarity condition shown for will be same as condition of invertibility for MA(2) given that polynomial is the similar, that is

$$\begin{cases} |\theta_2| < 1 \\ \theta_2 + \theta_1 < 1 \\ \theta_2 - \theta_1 < 1 \end{cases}$$

Example 2.6.1. Do As an exercise

- Show it is stationary (by direct calculation of $\mathbb{E}\left[X_t\right]$)
- Derive and plot the ACF, γ_0, γ_k (expect γ_k becomes 0 after second lag)
- Derive the Wold representation ($\psi_1 = -\theta_1, \psi_2 = -\theta_2, \psi_0 = 1, \psi_j = 0, \forall j > 2$)
- State (and show) the invertibility conditions

Chapter 3

Prediction

Here we use linearity of stationary processes in order to start making prediction of X_t given past values: this is based on *conditioning* and *conditional expectation* will be our predictor (best predictor in the MSE sense).

3.1 Premises

Example 3.1.1 (ex 1.1 ITSF)**.** Let X and Y be two random variables with $\mathbb{E}[Y] = \mu$ and $\mathbb{E}[Y^2] < \infty$

1. show that the constant c that minimizes $\mathbb{E}[Y - c]$ is $c = \mu$
2. deduce that the random variable $f(X)$ that minimizes $\mathbb{E}\left[(Y - f(X))^2 | X\right]$ is $f(X) = \mathbb{E}[Y|X]$.
3. deduce that the random variable $f(X)$ that minimizes $\mathbb{E}\left[(Y - f(X))^2\right]$ (no conditioning) is also $f(X) = \mathbb{E}[Y|X]$.

In order

1. to prove $\min_{c\in\mathbb{R}} \mathbb{E}\left[(Y - c)^2\right] = \mu$ the objective function to be minimized is

$$\begin{aligned}\phi(c) &= \mathbb{E}\left[Y^2 - 2cY + c^2\right] = \mathbb{E}\left[Y^2\right] - 2c\,\mathbb{E}[Y] + \mathbb{E}\left[c^2\right] \\ &= \mathbb{E}\left[Y^2\right] - 2c\mu + c^2\end{aligned}$$

   It's derivative with respect to c is

$$\frac{\partial}{\partial c}\phi(c) = -2\mu + 2c$$

   and equating to 0

$$\frac{\partial}{\partial c}\phi(c) = 0 \implies c = \mu$$

2. to deduce that the rv $f(X)$ (with f measurable function) that minimize $\mathbb{E}\left[(Y - f(X))^2 | X\right]$ (where the conditioning means the information/sigma

algebra $\sigma(X)$) is $\mathbb{E}[Y|X]$, the objective function to be minimized

$$\begin{aligned}\phi(f(X)) &= \mathbb{E}\left[Y^2 - 2f(X)Y + f^2(X)|X\right] \\ &= \mathbb{E}\left[Y^2|X\right] - 2\,\mathbb{E}\left[f(X)\cdot Y|X\right] + \mathbb{E}\left[f^2(X)|X\right] \\ &= \mathbb{E}\left[Y^2|X\right] - 2f(X)\cdot\mathbb{E}\left[Y|X\right] + f^2(X)\end{aligned}$$

The derivative with respect to $f(X)$

$$\frac{\partial}{\partial f(X)}\phi(f(X)) = -2\,\mathbb{E}[Y|X] + 2f(X)$$

and equating to zero we have $f(X) = \mathbb{E}[Y|X]$.
Hence the conditional expectation of Y given X is the function of X that minimises the square discance between Y and $f(X)$ (on average, conditional to X ..., minimizes also unconditionally)

3. todo based on the *law of iterated expectation will se*

Example 3.1.2 (Ex 1.2 ITSF)**.** Generaliziation of the problem to all the past

Proposition 3.1.1. *Two useful results on conditioning the first of which is the* law of iterated expectations *(l.i.e.)*

$$\begin{aligned}\mathbb{E}[Y] &= \mathbb{E}[\mathbb{E}[Y|X]] \\ \mathrm{Var}[Y] &= \mathbb{E}[\mathrm{Var}[Y|X]] + \mathrm{Var}[\mathbb{E}[Y|X]]\end{aligned}$$

Remark 19. For the law of iterated expectations, he unconditional expectation of the conditional expectation of Y given X is equal to the unconditional expectation of Y Here

- conditional means: based on / given some information (it will be natural for us to assume that the *information* is the *past* in the context of time series).[1]
- unconditional: no information available (one condition on the trivial sigma algebra)

Law of iterated expectation. If we have two rv X, Y and som mild condition about

$$\mathbb{E}[\mathbb{E}[Y|X]] = \mathbb{E}[Y]$$

where $\mathbb{E}[Y|X]$ is a measurable function of X, the conditioning variable.

$$\begin{aligned}\mathbb{E}[\mathbb{E}[Y|X]] &\overset{(1)}{=} \int_{R_X} \mathbb{E}[Y|X]\cdot f_X(x)\,\mathrm{d}x \\ &\overset{(2)}{=} \int_{R_X}\int_{R_Y} y\cdot f_{Y|X}(y|x)\,\mathrm{d}y\cdot f_X(x)\,\mathrm{d}x \\ &\overset{(3)}{=} \int_{R_Y} y\int_{R_X} f_{XY}(x,y)\,\mathrm{d}x\,\mathrm{d}y \\ &\overset{(4)}{=} \int_{R_Y} y\cdot f_Y(y)\,\mathrm{d}y \\ &= \mathbb{E}[Y]\end{aligned}$$

[1] The concept of information is crucial in general: for example with the fisher information the biggest the information is the smaller the variance of an unbiased estimation

where

- in (1) the outside expectation in $\mathbb{E}[\mathbb{E}[Y|X]]$ is with respect to X, so we'll have an integral with respect to the domain of X (we suppose $R_X = \mathbb{R}, R_Y = \mathbb{R}$)
- (2) we develop the inside integral attached to $\mathbb{E}[Y|X]$ that is
- (3) we assume we can move a little bit of quantities and especially that the product of $f_{Y|X}(y|x)$ for $f_X(x)$ is equal to the joint density
- (4) integrating out the joint we have the marginal

□

Proof. The other proof of variance as exercise □

3.2 Prediction

Definition 3.2.1 (Predictor)**.** It's defined as conditional expectation of tomorrow's rv given all the past

$$X_{n+1|n} = \mathbb{E}[X_{n+1}|X_1, \ldots, X_n]$$

Important remark 9. The predictor according to what we've seen (ex 1.1 ITSF) is the best predictor in the sense that minimises square distance between future value X_{n+1} and its prediction $X_{n+1|n}$ given the past information (conditional MSE) which is the loss function we want to minimize in making predictions

$$\mathbb{E}\left[(X_{n+1} - X_{n+1|n})^2 | X_1, \ldots, X_n\right]$$

and by the law of iterated expectation (l.i.e) also minimises the unconditional MSE

$$\mathbb{E}\left[(X_{n+1} - X_{n+1|n})^2\right]$$

ie $X_{n+1|n}$ is the minimum mean square error predictor of X_{n+1} given the past information set (sigma algebra generated by $X_1, \ldots, X_n$, called filtration)

$$\mathscr{F}_n = \{X_1, X_2, \ldots, X_n\}$$

Since by the l.i.e

$$\mathbb{E}\left[X_{n+1} - X_{n+x|n}\right] = 0$$

that is the predictor is *unbiased* then the minimum MSE is also the minimum variance, that is $X_{n+1|n}$ also minimises the *prediction error variance*:

$$\mathbb{E}\left[(X_{n+1|n} - X_{n+1})^2\right]$$

where the quantity $X_{n+1|n} - X_{n+1}$ is the *prediction error*

Remark 20. The predictor error variance is an important quantity in time series

Example 3.2.1 (One step ahead prediction error variance of $X_{n+1|n}$ when X is a linear process)**.** If $\{X_t\}_{t\in T}$ is a linear process that is

$$X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \quad \text{where } \psi_0 = 1, \sum_{j=0}^{\infty} \psi_j^2 < \infty, \varepsilon_t \sim \mathrm{WN}\left(0, \sigma^2\right)$$

The predictor is developed as follow

$$\begin{aligned}
X_{t+1|t} &= \mathbb{E}\left[X_{t+1}|X_t, X_{t-1}, \dots, X_1\right] \\
&= \mathbb{E}\left[\sum_{j=0}^{\infty} \psi_j \varepsilon_{t+1-j} | X_t, \dots, X_1\right] \\
&= \mathbb{E}\left[\varepsilon_{t+1} + \psi_1 \varepsilon_t + \psi_2 \varepsilon_{t-1} + \dots | \underbrace{X_t, \dots, X_1}_{\mathscr{F}_t}\right] \\
&= \mathbb{E}\left[\varepsilon_{t+1}|\mathscr{F}_t\right] + \psi_1 \mathbb{E}\left[\varepsilon_t|\mathscr{F}_t\right] + \psi_2 \mathbb{E}\left[\varepsilon_{t-1}|\mathscr{F}_t\right] + \dots \\
&= \mathbb{E}\left[\varepsilon_{t+1}\right] + \psi_1 \varepsilon_t + \psi_2 \varepsilon_{t-1} + \dots
\end{aligned}$$

where in the last passage

- $\mathbb{E}\left[\varepsilon_{t+1}|\mathscr{F}_t\right] = \mathbb{E}\left[\varepsilon_{t+1}\right]$ because ε_{t+1} is uncorrelated with past observation
- if X_t is measurable with respect to $\mathscr{F}_t$ the same occurs to the ε_t and so these are taken out of conditional expectation

The last equation is our predictor: basically we add the expected value

Now the **prediction error** is the difference between future and our the prediction

$$X_{t+1} - X_{t+1|t}$$

It's expected value given the past

$$\begin{aligned}
\mathbb{E}\left[X_{t+1} - X_{t+1|t}|\mathscr{F}_t\right] &= \mathbb{E}\left[X_{t+1}|\mathscr{F}_t\right] - \mathbb{E}\left[X_{t+1|t}|\mathscr{F}_t\right] \\
&\overset{(1)}{=} X_{t+1|t} - X_{t+1|t} \\
&= 0
\end{aligned}$$

where in (1) $\mathbb{E}\left[X_{t+1}|\mathscr{F}_t\right] = X_{t+1|t}$ by definition of the predictor while $\mathbb{E}\left[X_{t+1|t}|\mathscr{F}_t\right] = X_{t+1|t}$ by construction (in the sense that being a function of past variable up to t so is known at time t and thus we can get it out of expectation).
So the prediction error has null expected value
If we take the unconditional expectation

$$\mathbb{E}\left[\mathbb{E}\left[X_{t+1} - X_{t+1|t}\right]|\mathscr{F}_t\right] = \mathbb{E}\left[0\right] = 0$$

The prediction error variance (for a prediction error which is unbiased/has zero mean)

$$\begin{aligned}
\mathbb{E}\left[(X_{t+1} - X_{t+1|t})^2|\mathscr{F}_t\right] &= \mathbb{E}\left[\left(\sum_{j=0}^{\infty} \psi_j \varepsilon_{t+1-j} - \sum_{j=1}^{\infty} \psi_j \varepsilon_{t+1-j}\right)^2 |\mathscr{F}_t\right] \\
&= \mathbb{E}\left[\varepsilon_{t+1}^2|\mathscr{F}_t\right] = \mathbb{E}\left[\varepsilon_{t+1}^2\right] \\
&= \sigma^2
\end{aligned}$$

This σ^2 is the prediction error variance.

Summarizing we have

- $X_1, \dots, X_m$ past variable which we call $\mathscr{F}_n$
- predictor $X_{n+1|n} = \mathbb{E}[X_{n+1}|X_1, \dots, X_n]$ which is best in the sense that minimise the meas square error
- prediction error is $X_{n+1|n} - X_{n+1}$: it has conditional and unconditional zero mean, $\mathbb{E}\left[X_{n+1|n} - X_{n+1}|\mathscr{F}_n\right] = 0$ and $\mathbb{E}\left[X_{n+1|n} - X_{n+1}\right] = 0$
- **prediction error variance** $\mathbb{E}\left[(X_{n+1|n} - X_{n+1})^2\right]$.
  For a linear process it's always equal to σ^2 and it's **smaller than the unconditional variance** $\gamma_0 = \sigma^2 \sum_{j=0}^{\infty} \psi_j \psi_j$:
  - eg for AR(1) $\gamma_0 = \frac{\sigma^2}{1-\psi^2} > \sigma^2$ (with $|\phi| < 1$)
  - for MA(1) $\gamma_0 = \sigma^2(1 + \theta^2) > \sigma^2$

  So having information on the past implies smaller variability

3.3 Best linear predictors

Previously we seen that

$$X_{n+1|n} = \mathbb{E}[X_{n+1}|X_n, \dots, X_1]$$

is the best predictor. Now we want to restrict/focus on the best *linear* predictor

$$X_{n+1|n} = \mathbb{E}[X_{n+1}|X_n, \dots, X_1] = \phi_{n1}X_n + \phi_{n2}X_{n-1} + \ldots + \phi_{nn}X_1$$

which is a linear combination of past observation with coefficient $\phi_{n1}, \dots, \phi_{nn}$ such that the distance between future value and linear predictor

$$\mathbb{E}\left[(X_{n+1} - (\phi_{n1}X_n + \phi_{n2}X_{n-1} + \ldots + \phi_{nn}X_1))^2|\mathscr{F}_n\right]$$

is minimum. THe minimization can be carried out (as in previous class) algebrically by derive wrt to all the parameters and equating to 0

$$\frac{\partial}{\partial\phi_{n1}}\frac{\partial}{\partial\phi_{n2}} \cdots \frac{\partial}{\partial\phi_{nn}} f() = 0$$

with f some kind of objective function.
Alternatively, equivalently we use a geometric interpretation/method

$$\mathbb{E}\left[(X_{n+1} - X_{n+1|n})X_j\right] = 0, \quad \forall j = 1, \dots, n$$

ie $X_{n+1|t}$ is the orthogonal projection of X_{n+1} onto the (infinite Hilbert) vector space spanned by $X_1, \dots, X_n$. That is we want to find coefficients

$$X_{n+1|n} = \phi_{n1}X_n + \phi_{n2}X_{n-1} + \ldots + \phi_{nn}x_1$$

such that is minimum the following quantity

$$\mathbb{E}\left[(X_{n+1}-X_{n+1|n})^2\right]$$

We can think of a vector space created by the past values $\mathcal{C} = span\,\{X_1,\dots,X_n\}$ (∞-dimensional hilbert space) basically it's a vector space of function but we can think iti); we have that the value of tomorrow does not belong to this vector space ($X_{n+1} \notin \mathcal{C}$) otherwise the value of tomorrow would be just a linear combination of values up to today. We look for a vector in $X_{n+1|n} \in \mathcal{C}$ (so which can thus be obtained by linear combination of past values) which minimizes the (squared) distance between the prediction (our vector) and the future real observation X_{n+1}. To have it with minimum distance we have to project the future observation orthogonally on our $\mathcal{C}$ and find the corresponding vector.
All the vector belonging to $\mathcal{C}$ are *orthogonal* to the *distance* between future value and its projection; thus even each single past value should be orthogonal (set the coefficient of the other elements of the basis to 0) with inner product with distance being 0:

$$\left\langle X_j,(X_{n+1}-X_{n+1|n})\right\rangle = 0, \quad \forall j=1,\dots,n$$

In this space the inner product is represented by the expectation of the product of the two vectors (basically it's saying rv with inner product null aren't correlated i guess). So this above is equivalent to having

$$\mathbb{E}\left[(X_{n+1}-X_{n+1|n})\cdot X_j\right] = 0, \quad \forall j=1,\dots,n$$

Which are the implication here? Let's consider the case $j=n$ (assume process stationary with $\mathbb{E}[X_t]=0, \forall t$, with $\gamma_k=\gamma_{-k}$)

$$\begin{aligned}
&\mathbb{E}\left[(X_{n+1}-X_{n+1|n})X_n\right] = 0\\
&\mathbb{E}\left[(X_{n+1}-(\phi_{n1}X_n+\phi_{n2}X_{n-1}+\ldots+\phi_{nn}X_1))\cdot X_n\right] = 0\\
&\underbrace{\mathbb{E}\left[X_{n+1}\cdot X_n\right]}_{\gamma_1} = \phi_{n1}\,\mathbb{E}\left[X_nX_n\right]+\phi_{n2}\,\mathbb{E}\left[X_nX_{n-1}\right]\ldots+\phi_{nn}\,\mathbb{E}\left[X_1\cdot X_n\right]\\
&\gamma_1 = \phi_{n1}\gamma_0+\phi_{n2}\gamma_1+\ldots+\phi_{nn}\gamma_{n-1}
\end{aligned}$$

If $j=n-1$ developing the same way and starting from the last point we have

$$\begin{aligned}
\mathbb{E}\left[X_{n+1}X_n\right] &= \phi_{n1}\,\mathbb{E}\left[X_nX_{n-1}\right]+\ldots+\phi_{nn}\,\mathbb{E}\left[X_1X_{n-1}\right]\\
\gamma_2 &= \phi_{n1}\gamma_1+\phi_{n2}\gamma_0+\ldots+\phi_{nn}\gamma_{n-2}
\end{aligned}$$

We can go ahead up to $j=1$ and putting everything together, the above (orthogonality or least squares) condition is equivalent to:

$$\begin{aligned}
\begin{bmatrix}\gamma_1\\ \gamma_2\\ \cdots\\ \gamma_n\end{bmatrix} &= \begin{bmatrix}\gamma_0 & \gamma_1 & \gamma_2 & \cdots & \gamma_{n-1}\\ \gamma_1 & \gamma_0 & \gamma_1 & \cdots & \gamma_{n-2}\\ \gamma_2 & \gamma_1 & \gamma_0 & \cdots & \gamma_{n-3}\\ \vdots & & & & \\ \gamma_{n-1} & \gamma_{n-2} & \gamma_{n-3} & \cdots & \gamma_0\end{bmatrix}\begin{bmatrix}\phi_{n1}\\ \phi_{n2}\\ \cdots\\ \phi_{nn}\end{bmatrix}\\
\boldsymbol{\gamma}_n &= \boldsymbol{\Gamma}_n\boldsymbol{\phi}_n
\end{aligned}$$

where

- $\boldsymbol{\gamma}_n$ are the autocovariances of X_t for $k = 1, \ldots, n$
- $\boldsymbol{\Gamma}_n$ is the so called Toeplitz matrix associate with this system, ie the variances-autocovariances matrix of $\mathbf{X} = [X_n, X_{n-1}, \ldots, X_1]^\intercal$, $\boldsymbol{\Gamma}_n$ is a $n \times n$ matrix defined with $\boldsymbol{\Gamma}_n = \mathbb{E}\left[\mathbf{X}\mathbf{X}^\intercal\right]$ (assumin the vector is zero meaned), the main diagonal is composed of variance (constant) and symmetric covariance out of diagonal

Again we want to find ϕ_n which are the coefficient of the best linear predictor we're searching to obtain by minimization; the coefficients can be obtained by solving the linear system

$$\boldsymbol{\phi}_n = \boldsymbol{\Gamma}_n^{-1}\boldsymbol{\gamma}_n \tag{3.1}$$

Solving the system can be done by inverting $\boldsymbol{\Gamma}_n$ (this can be done whenever the process is stationary); we can divide both member of equation above by γ_0 obtaining something related to autocorrelation

$$\frac{1}{\gamma_0}\boldsymbol{\gamma}_n = \frac{1}{\gamma_0}\boldsymbol{\Gamma}_n\boldsymbol{\phi}_n$$
$$\begin{bmatrix}\rho_1\\ \rho_2\\ \cdots\\ \rho_n\end{bmatrix} = \begin{bmatrix}1 & \rho_1 & \cdots & \cdots & \rho_{n-1}\\ \vdots & & & & \\ \vdots & & & & \\ \vdots & & & & \\ \rho_{n-1} & \rho_{n-2} & \rho_{n-3} & \cdots & 1\end{bmatrix}\begin{bmatrix}\phi_{n1}\\ \phi_{n2}\\ \cdots\\ \phi_{nn}\end{bmatrix}$$

then invert this last autocorrelation matrix to obtain the coefficient in the equivalent system.

Important remark 10 (Practical). The best linear predictor $\hat{x}_{n+1|n}$ of X_{n+1} given the past information set can be then obtained as follows:

- given an observed stationary time series $x_1, \ldots, x_n$, estimate the autocovariance function as

  $$\widehat{\gamma}_k = \frac{1}{n}\sum_{t=1}^{n-k}(x_t - \overline{x}_b)(x_{t+k} - \overline{x}_n)$$

  where

  - note we're using n as denominator which is biased (the sample acv should have $n - k$ instead)
  - the quantity

    $$\overline{x}_n = \frac{1}{n}\sum_{t=1}^{n} x_t$$

    is the sample mean of the serie

- construct the sample vector $\hat{\gamma_n}$ and the sample covariance matrix $\hat{\boldsymbol{\Gamma}_n}$ to obtain the estimated coefficients

  $$\hat{\phi}_n = \hat{\boldsymbol{\Gamma}}_n^{-1}\hat{\gamma}_n$$

- obtain

$$\hat{x}_{n+1|n} = \hat{\phi}_{n1}x_n + \hat{\phi}_{n2}x_{n-1} + \dots \hat{\phi}_{nn}x_1$$

Moreover we can obtain an estimation of σ^2 (variance of the prediction error)

$$\begin{aligned}\hat{\sigma}^2 &= \mathbb{E}\left[(X_{n+1} - X_{n+1|n})^2\right] = \mathbb{E}\left[X_{n+1}^2 - 2\boldsymbol{\phi}_n^\mathsf{T}\boldsymbol{X}_n\mathbf{X}_{n+1} + (\boldsymbol{\phi}_n^\mathsf{T}\mathbf{X}_n)^2\right]\\ &= \gamma_0 - 2\boldsymbol{\phi}_n^\mathsf{T}\boldsymbol{\gamma}_n + \mathbb{E}\left[\boldsymbol{\phi}_n^\mathsf{T}\boldsymbol{X}_n\boldsymbol{X}_n^\mathsf{T}\boldsymbol{\phi}_n\right]\\ &= \gamma_0 - 2\boldsymbol{\phi}_n^\mathsf{T}\boldsymbol{\gamma}_n + \boldsymbol{\phi}_n^\mathsf{T}\boldsymbol{\Gamma}_n\boldsymbol{\phi}_n\\ &= \hat{\gamma}_0 - \hat{\boldsymbol{\phi}}_n^\mathsf{T}\hat{\boldsymbol{\gamma}}_n\end{aligned}$$

which follows by 3.1.

Example 3.3.1 (The best linear predictor for an AR(1) process (ex 2.5.1 ITSF))**.** if

$$X_t = \omega + \phi X_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \text{WN}\left(0, \sigma^2\right)$$

with $\mathbb{E}\left[X_t\right] = \frac{w}{1-\phi}$ if $\omega = 0$ it's zeromean ar(1) stationary if $|\phi| < 1$ and $\rho_k = \phi^{|k|}$.
Construct the best linear predictor based on the formula seen before (here we use the autocorrelation version/ premultiplied by $\frac{1}{\gamma_0}$)

$$\begin{aligned}\boldsymbol{\gamma}_n &= \Gamma_n\boldsymbol{\phi}_n\\ \gamma_n\frac{1}{\gamma_0} &= \frac{1}{\gamma_0}\Gamma_n\phi_n\end{aligned}$$

Doing calculation and remembering for AR(1) $\rho_k = \phi^{|k|}$ the system becomes

$$\begin{bmatrix}\phi\\ \phi^2\\ \dots\\ \phi^n\end{bmatrix} = \begin{bmatrix}1 & \phi & \phi^2 & \dots & \phi^{n-1}\\ \phi & \phi^2 & \phi & \dots & \phi^{n-2}\\ \dots & & & & \\ \phi^{n-1} & \phi^{n-2} & \dots & \dots & 1\end{bmatrix}\begin{bmatrix}\phi_{n1}\\ \phi_{n2}\\ \dots\\ \phi_{nn}\end{bmatrix}$$

We observe that a possible solution is the vector (equality above is satisfied with following substitution)

$$\begin{bmatrix}\phi_{n1}\\ \phi_{n2}\\ \dots\\ \phi_{n1}\end{bmatrix} = \begin{bmatrix}\phi\\ 0\\ \dots\\ 0\end{bmatrix}$$

So the best linear prodictor for AR(1)

$$\begin{aligned}X_{n+1|n} &= \phi_{n1}X_n + \phi_{n2}X_{n-1} + \dots \phi_{nn}X_1\\ &= \phi X_n + 0X_{n-1} + \dots 0X_1\\ &= \phi X_n\end{aligned}$$

This is somehow expected since in AR(1) with $\omega = 0$

$$X_t = \phi X_{t-1} + \varepsilon_t, \quad |\phi| < 1, \varepsilon_t \in \text{WN}\left(0, \sigma^2\right)$$

if we take the conditional expectation (the best overall predictor)

$$
\begin{aligned}
X_{n+1|n} &= \mathbb{E}\left[X_{n+1}|X_n, X_{n-1}, \ldots, X_1\right] \\
&= \mathbb{E}\left[\phi X_n + \varepsilon_{n+1}|X_n, X_{n-1}, \ldots, X_1\right] \\
&= \phi\,\mathbb{E}\left[X_n|X_n, X_{n-1}, \ldots, X_1\right] + \mathbb{E}\left[\varepsilon_{n+1}|X_n, X_{n-1}, \ldots, X_1\right] \\
&= \phi X_n + \mathbb{E}\left[\varepsilon_{n+1}\right] \\
&= \phi X_n
\end{aligned}
$$

So the above, which is the best linear predictor (MMSE) is also a linear predictor of $X_n, \ldots, X_1$ with weights $\phi, 0, \ldots, 0$, so it must be the best linear predictor (as developed above).
So in case of AR(1) the best predictor $X_{n+1|n} = \mathbb{E}\left[X_{n+1}|\mathscr{F}_n\right]$ is also the best *linear* predictor $X_{n+1|n} = \phi_{n1}X_n + \phi_{n2}X_{n-1} + \ldots + \phi_{nn}x_1 = \phi X_n$.
This is not always true, eg we could have a best predictor that is non linear (which is not)

Important remark 11 (MA(q)). Warning: for a MA(q) process

$$X_t = \varepsilon_t - \theta\varepsilon_{t-1}$$

(with ε_t being $\mathscr{F}_t$ measurable) to construct a predictor using $x_1, \ldots, x_n$ (which is what we observe)

$$
\begin{aligned}
x_t &= \varepsilon_t - \theta\varepsilon_{t-1} \\
\varepsilon_t &= x_t + \theta\varepsilon_{t-1} \\
&= x_t + \theta(x_{t-1} + \theta\varepsilon_{t-2}) \\
&= x_t + \theta x_{t-1} + \theta^2(x_{t-2} + \theta\varepsilon_{t-3}) \\
&= \sum_{j=0}^{\infty} \theta^j x_{t-j}
\end{aligned}
$$

in order to construct a predictor as a function of the observable variables

Chapter 4

Partial autocorrelation function

Remark 21. It's an useful tool to (among other) identify MA or AR process of order q/p from a stationary time series.
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Definition 4.0.1 (The formal definition)**.** the partial autocorrelation α_k is not definined for $k = 0$ (differently from the global autocorrelation seen so far which is)

$$\begin{aligned} a_1 &= \text{Corr}\left(X_2, X_1\right) = \rho_1 \\ a_k &= \text{Corr}\left(X_{k+1} - \mathbb{E}\left[X_{k+1}|X_2, \dots X_k\right], X_1 - \mathbb{E}\left[X_1, X_2, \dots X_k\right]\right) \end{aligned}$$

The peculiar is the last one where the partial autocorrelation for a lag of k is the autocorrelation between X_{k+1} and X_1 after we have subtracted from both each the conditional expectations/best prediction given $X_2, \dots X_k$, that are the variable between X_1 and X_{k+1}

Definition 4.0.2 (Equivalent definition)**.** A much more intuitive definition we tipically use is the following

$$a_k = \phi_{kk}$$

where

$$\mathbb{E}\left[X_{k+1}|X_1, \dots, X_k\right] = \phi_{k1}X_k + \phi_{k2}X_{k-1} + \dots \phi_{kk}X_1 +$$

Here basically is the regression coefficient of X_1 on the prediction/regression of X_{k+1} after having considered all the other times $X_2, \dots, X_k$.
If we take the derivative ϕ_{kk} is the increment to $\mathbb{E}\left[X_{k+1}|X1, \dots, X_k\right]$ following unity increase in X_1

Tool to calculate ϕ_{kk}: were we use the same techniques used to determine coefficient of the best linear predictor. We know that for the best linear predictor, the difference $X_{k+1} - \mathbb{E}\left[X_{k+1}|X_1, \dots, X_k\right]$ is *orthogonal* to all the vector spanning from $\{X_1, \dots, X_k\}$

$$X_{k+1} - \mathbb{E}\left[X_{k+1}|X_1, \dots, X_k\right] \perp\!\!\!\perp X_1, \dots, X_k$$

and thus for orthogonality what we have to have is that the inner product

$$\mathbb{E}\left[X_{k+1} - \phi_{k1}X_k - \phi_{k2}X_{k-1} - \ldots - \phi_{kk}X_1\right]X_j = 0 \quad \forall j = 1, \ldots, k$$

Taking expectations and dividing both the members by γ_0

$$\rho_{k+1-j} = \phi_{k1}\rho_{k-j} + \phi_{k2}\rho_{k-1-j} + \ldots \phi_{kk}\rho_{1-j}, \quad j = 1, \ldots, k$$

In matrix form

$$\underbrace{\begin{bmatrix} \rho_1 \\ \rho_2 \\ \cdots \\ \rho_k \end{bmatrix}}_{\boldsymbol{\rho}_k} = \underbrace{\begin{bmatrix} \rho_0 & \rho_1 & \cdots & \rho_{k-1} \\ \rho_1 & \rho_0 & \cdots & \rho_{k-2} \\ \cdots & & & \\ \rho_{k-1} & \rho_{k-1} & \cdots & \rho_0 \end{bmatrix}}_{R_k} \underbrace{\begin{bmatrix} \phi_{k1} \\ \phi_{k2} \\ \cdots \\ \phi_{kk} \end{bmatrix}}_{\boldsymbol{\phi}_k}$$

and ϕ_{kk} can be obtained recursively or by the Cramer rule.
Here we have the same linear system as the one used to determine the coefficients of the BLP but differently here k does vary and $\forall k = 1, 2, \ldots$ there is a different system of dimension $k \times k$.
Note that here we only need $\phi_k k$; there is no need to solve the whole system but we can use cramer rule where, to have the last element of the vector ϕ_k we can do the ratio of determinant of following matrices

$$\phi_{kk} = \frac{\left|R_k^{(k)}\right|}{|R_k|}$$

with

- R_k as in the previous, the acf matrix associated with the process X_t. The determinant at the denominator is always different from 0 for stationary processes

- $R_k^{(k)}$ is R_k but with the last column replaced by the vector of known terms

$$R_k^{(k)} = \begin{bmatrix} \rho_0 & \rho_1 & \cdots & \rho_1 \\ \rho_1 & \rho_0 & \cdots & \rho_2 \\ \cdots & & & \\ \rho_{k-1} & \rho_{k-1} & \cdots & \rho_k \end{bmatrix}$$

Example 4.0.1 (AR(1) stationary process with zero mean)**.** With $X_t = \phi X_{t-1} + \varepsilon_t$, with $|\phi| < 1$ and $\varepsilon_t \sim \mathrm{WN}\left(0, \sigma^2\right)$ we have that

$$\rho_k = \phi^{|k|}, \quad k = 0, \pm 1, \pm 2, \ldots$$

and

$$\phi_{kk} = \begin{cases} \phi_{11} = \rho_1 = \phi & \text{if } k = 1 \\ \phi_{22} = \dfrac{\left|\begin{bmatrix} \rho_0 & \rho_1 \\ \rho_1 & \rho_2 \end{bmatrix}\right|}{\left|\begin{bmatrix} \rho_0 & \rho_1 \\ \rho_1 & \rho_0 \end{bmatrix}\right|} = \dfrac{\left|\begin{bmatrix} 1 & \phi \\ \phi & \phi^2 \end{bmatrix}\right|}{\left|\begin{bmatrix} 1 & \phi \\ \phi & 1 \end{bmatrix}\right|} = \frac{\phi^2-\phi^2}{1-\phi^2} = 0 & \text{if } k = 2 \\ \phi_{33} = \dfrac{\left|\begin{bmatrix} \rho_0 & \rho_1 & \rho_1 \\ \rho_1 & \rho_0 & \rho_2 \\ \rho_2 & \rho_1 & \rho_3 \end{bmatrix}\right|}{|R_3|} = \dfrac{\left|\begin{bmatrix} 1 & \phi & \phi \\ \phi & 1 & \phi^2 \\ \phi^2 & \phi & \phi^3 \end{bmatrix}\right|}{|R_3|} = \frac{0}{R_3} = 0 & \text{if } k = 3 \end{cases}$$

In general for an AR(1) process

$$\phi_{kk} = \begin{cases} \phi, & k = 1 \\ 0 & k > 1 \end{cases}$$

In general for an AR(p) processes PACF start in ϕ for $k = 1$ then has an decay and becomes zero after lag p. So we can use this to identify the order of the process.
WIth a mixed ARMA we can't use this because of the following

Example 4.0.2 (MA)**.** Derive $\phi_{kk}, k = 1, 2$ for a MA(1). (*it doesnt get to 0*)

Chapter 5

Sample stats

We're left with doing inference on the *moments* of a *stationary* stochastic process: we're interested

- estimation of parameters μ, γ_k and ρ_k

- deriving asymptotic distribution of sample mean $\overline{x}_n$ and correlation $\hat{\rho}_k$

- testing for serial correlation

of a stationary stochastic process. References are in ITSF chapter 2.
Other things that can be interesting are (to read if needed on slides)

5.1 Mean

5.1.1 Estimation

ASsuming we've observed the time series $x_1, x_2, \ldots, x_n$ coming from a stationary stochastic process with mean μ, variance γ_0 and covariance γ_k; we can estimate the sample mean is the moment estimator we will consider

$$\overline{x}_n = \frac{1}{n}\sum_{t=1}^{n} x_t$$

is an unbiased estimator for μ (if we take expectation of $\overline{x}_n$ we'll get μ *if the process is stationary*), the variance of $\overline{x}_n$ is

$$\mathrm{Var}\left[\overline{x}_n\right] = \mathbb{E}\left[(\overline{x}_n - \mu)^2\right]$$

suppose now that $\mu = 0$[1]. We can write the variance as (we're assuming that $\mathbb{E}[X_t] = 0$)

$$\begin{aligned}
\operatorname{Var}[\overline{x}_n] &= \mathbb{E}\left[\left(\frac{1}{n}\sum_{t=1}^{n} x_t\right)\left(\frac{1}{n}\sum_{s=1}^{n} x_s\right)\right] = \frac{1}{n^2}\,\mathbb{E}\left[\left(\sum_{t=1}^{n} x_t\right)\left(\sum_{s=1}^{n} x_s\right)\right] \\
&\overset{(1)}{=} \frac{1}{n^2}(n\gamma_0 + 2(n-1)\gamma_1 + 2(n-2)\gamma_2 + \dots 2\gamma_{n-1}) = \frac{1}{n}\left(\gamma_0 + 2\sum_{k=1}^{n}\frac{n-k}{n}\gamma_k\right) \\
&= \frac{1}{n}\sum_{|k|<n}\frac{n-|k|}{n}\gamma_k
\end{aligned}$$

with (1) coming from basic algebra.
This above is the variance of the sample mean estimator

5.1.2 Ergodicity of $\overline{x}_n$

In general ergodicity is a property of an estimator to converge to the actual moment.
Here we want to prove that $\overline{x}_n \to \mu$:

- almost surely: ergodic theorem for strongly stationary processes
- or in means square (ergodic theorem for weakly or second order stationary processes).

These are law of large numbers for dependent data, which require a resttriction on the memory of the process. One of these restrictions is ergodicity, sort of average asymptotic independence

Definition 5.1.1. A weakly stationary stochastic process $\{X_t\}_{t\in T}$ is ergodic with respect to moment $\mathbb{E}[g(X_t)]$ if the sample moment $\frac{1}{n}\sum_{t=1}^{n} g(x_t)$ converges (in probability or almost surely) to $\mathbb{E}[g(X_t)]$ that is

$$\frac{1}{n}\sum_{t=1}^{n} g(x_t) \to \mathbb{E}[g(X_t)]$$

Remark 22. As we said $\{X_t\}_{t\in T}$ ergodic with resepect to the mean if

$$\frac{1}{n}\sum_{t=1}^{n} x_t \to \mu$$

Important remark 12. The property of ergodicity ensures the *consistency* of estimators of the moments.

Theorem 5.1.1. *If $\{X_t\}_{t\in T}$ is stationary with mean μ and covariance γ_k then as $n \to \infty$*

$$\begin{aligned}
\operatorname{Var}[\overline{x}_n] &= \mathbb{E}\left[(\overline{x}_n - \mu)^2\right] \to 0 \quad \textit{if } \gamma_n \to 0 \\
n\operatorname{Var}[\overline{x}_n] &\to \sum_{k=-\infty}^{\infty} \gamma_k = 2\pi f(0) \quad \textit{if } \sum_{k=-\infty}^{\infty} |\gamma_k| < \infty
\end{aligned}$$

[1] Remind that even though the mean of a process is zero we shall be going to estimate it.

Proof. Dimostrazione di $\mathrm{Var}[\overline{x}_n] \to 0$ as $n \to \infty$ Pagina 10 di 14 haandout 7 sample mean □

Remark 23. Thus the variance of the sample mean converges to 0 if the covariance at lag n converges to zero.
In large sample by multiplying per n converges to the sum of autocovariances from $-\infty$ to $+\infty$ (if the sequence of autocovariances is absolutely summable), where f is the spectral density evaluated in zero

Important remark 13. if $\{x_t\}_{t\in T}$ is a linear process then the variance of sample mean multiplied by n does converge to $\sigma^2\left(\sum_{j=0}^{\infty}\psi_j\right)^2$

$$n\,\mathrm{Var}[\overline{x}_n] \to \sum_{k=-\infty}^{\infty}\gamma_k = 2\pi f(0) = \sigma^2\left(\sum_{j=0}^{\infty}\psi_j\right)^2$$

Theorem 5.1.2. *If $\{X_t\}_{t\in T}$ is the linear process $X_t = \mu + \sum_{j=0}^{\infty}\psi_j\varepsilon_{t-j}$ with $\sum_{j=0}^{\infty}|\psi_j| < \infty$, $\varepsilon_t \sim IID(0,\sigma^2)$ then*

$$\sqrt{n}(\overline{x}_n - \mu) \xrightarrow{d} \mathrm{N}(0, v)$$

where

$$v = \sum_{k=-\infty}^{\infty}\gamma_k = 2\pi f(0) = \sigma^2\left(\sum_{j=0}^{\infty}\psi_j^2\right)$$

Important remark 14. If the linear process $\{x_t\}_{t\in T}$ is also gaussian then

$$\sqrt{n}(\overline{x_n} - \mu) \sim \mathrm{N}\left(0, \sum_{|k|<n}\frac{n-|k|}{n}\gamma_k\right)$$

Remark 24. Based on this we are able to construct asymptotic interval for the sample mean estimator of a stationary time series

5.2 Autocovariance and autocorrelation

For the estimate

$$\hat{\gamma}_k = \frac{1}{n}\sum_{t=1}^{n-k}(x_t - \overline{x}_n)(x_{t+k} - \overline{x}_n)$$

and

$$\hat{\rho}_k = \frac{\hat{\gamma}_k}{\hat{\gamma}_0}$$

Notice that $\hat{\rho}_k = \hat{\rho}_{-k}$

Important remark 15. $\hat{\gamma}_k$is biased for γ_k but asymptotically unbiased and more efficient than the unbiased estimator obtained by dividing by $n-k$ rather by n

Theorem 5.2.1. *If $\{x_t\}_{t\in T}$ is the linear process $x_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$, with $\sum_{j=0}^{\infty} |\psi_j| < \infty$, $\varepsilon_t \sim IID(0, \sigma^2)$, and $\mathbb{E}\left[x_t^4\right] < \infty$ then*

$$\sqrt{n}(\hat{\rho}(h) - \rho(h)) \xrightarrow{d} \mathrm{N}\left(0, W\right)$$

where

$$\hat{\rho}(h) = [\hat{\rho}_1, \hat{\rho}_2, \ldots, \hat{\rho}_h]^\mathsf{T}$$
$$\rho(h) = [\rho_1, \rho_2, \ldots, \rho_h]^\mathsf{T}$$

and W has generic element (Bartlett formula), with $i, j = 1, \ldots, h$

$$w_{ij} = \sum_{k=-\infty}^{\infty} (\rho_{k+i}\rho_{k+j} + \rho_{k-i}\rho_{k+j} + 2\rho_i rho_j rho_k^2 - 2rho_i rho_k rho_{k+j} - 2\rho_j\rho_k\rho_{k+i})$$

Important remark 16. straightforward algebra shows that

$$w_{ij} = \sum_{k=-\infty}^{\infty} (rho_{k+i} + rho_{k-i} - 2\rho_i\rho_k)(rho_{k+j} + rho_{k-j} - 2\rho_j\rho_k)$$

5.2.1 Bartlett formula for a white noise process

If $\{x_t\}$ is a white noise process then

$$w_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

and an approximate confidence interval for ρ_k is

$$I_{0.95}(\rho) = \left(-1.96\frac{1}{\sqrt{n}}, 1.96\frac{1}{\sqrt{n}}\right)$$

This can be used as a check for the significance of the single estimated autocorrelation coefficients $\hat{\rho}_k$

5.2.2 Testing the estimated noise sequence

These are all statistical test which test if a group of autocorrelations of a time series are different from zero.[2]. The hypotheses are:

- H_0: The data is **not correlated** (i.e. the correlations in the population from which the sample is taken are 0, so that any observed correlations in the data result from randomness of the sampling process).

- H_1: The data exhibit serial correlation

[2]Instead of testing randomness at each distinct lag, it tests the "overall" randomness based on a number of lags, and is therefore a **portmanteau test**

1. the Box-Pierce: under H_0

$$Q_{BP} = n \sum_{j=1}^{h} \hat{\rho}(j)^2 \sim \chi^2(h)$$

The Box–Pierce test statistic is a simplified version of the Ljung–Box statistic for which subsequent simulation studies have shown poor performance [3] so the following has to be preferred

2. **Ljung-box test**

$$Q_{LB} = n(n+2) \sum_{j=1}^{h} \frac{\hat{\rho}(j)^2}{n-j} \sim \chi^2(h)$$

where n is the sample size $\rho(j)$ is the sample autocorrelation at lag j and h is the *number of lags being tested*. The Box-Ljung is an omnibus test of independence at all lags up to the one you specify. Hyndiman and Athanasopoulos in their forecasting book[4] recommend using as h:

- $h = 10$ for non seasonal data
- $h = 2m$ for seasonal data, where m is the period of seasonality

Under H_0 the statistics follow asymptotically follows $\chi^2_{(h)}$. For significance level α , the critical region for rejection of the hypothesis of randomness is $Q > \chi^2_{1-\alpha,h}$ where $\chi^2_{1-\alpha,h}$ is the $(1-\alpha)$-quantile of the chi-squared distribution with h degrees of freedom.
The Ljung-Box test is widely applied in econometrics and other applications of time series analysis. A similar assessment can be also carried out with the Breusch-Godfrey test and the Durbin–Watson test.

3. MCleod Li

$$Q_{ML} = n(n+2) \sum_{j=1}^{h} \frac{\hat{\rho}_{y^2}(j)^2}{n-j} \sim \chi^2(h)$$

5.3 Higher order moment estimation

Definition 5.3.1 (Skewness). is

$$S = \frac{\mathbb{E}\left[(y_t - \mu)^3\right]}{\sigma^3}$$

with moment estimator

$$\hat{S} = \frac{\frac{1}{n}\sum_{t=1}^{n}(y_t - \overline{y})^3}{\hat{\sigma}^3}$$

where under H_0 $\hat{\sigma}^2 = \sqrt{\gamma 0}$

$$\sqrt{n}(\hat{S}) \xrightarrow{d} \mathrm{N}\left(0, 6\right)$$

[3] Ljung-Box statistic is closer to a $\chi^2(h)$ distribution than is the distribution for the Box-Pierce statistic for all sample sizes. See https://en.wikipedia.org/wiki/Ljung%E2%80%93Box_test

[4] See https://robjhyndman.com/hyndsight/ljung-box-test/

Definition 5.3.2 (Kurtosis)**.** THe kurtosis and the excess kurtosis $(K-3)$

$$K = \frac{\mathbb{E}\left[(y_t - \mu)^4\right]}{\sigma^4}$$

moment estimator

$$\hat{K} = \frac{\frac{1}{n}\sum_{t=1}^{n}(y_t - \overline{y})^4}{\hat{\sigma}^4}$$

finally

$$\sqrt{n}(\hat{K} - 3) \xrightarrow{d} \mathrm{N}\left(0, 24\right)$$

5.4 Testing normality

Jarque bera test for normality

$$BJ = n\left(\frac{\hat{S}^2}{6} + \frac{(\hat{K}-3)^2}{24}\right) \sim_{H_0} \chi^2(2)$$

Chapter 6

Transformations

6.1 Stationarity in mean

How to make a series stationary in mean

- Estimate a polynomial trend with a linear filter (weighted average) and then subtract it from the series
- Use an exponential smoothing filter
- Apply an high pass filter (designed in the frequency domain)
- Apply a difference operator (for deterministic and stochastic trends): Differencing $(1-L)^d X_t$ stationary process

6.1.1 Linear filters for trend estimation

Let

$$X_t = m_t + Z_t$$

where m_t trend component and Z_t stationary zero mean stochastic process. We are interested in estimating m_t .

- for *linear trends and at the interior of the series*, simple 2-sided moving average

$$\hat{m}_t = \frac{1}{2h+1} \sum_{j=-h}^{h} X_{t-j}$$

- weighted averages

$$\hat{m}_t = \sum_{j=-h}^{h} w_j X_{t-j}$$

λ	Transformation
-2	$1/x^2$
-1	$1/x$
-0.5	$1/\sqrt{x}$
0	$\log x$
0.5	$\sqrt{x}$
1	x (no trasformation)
2	x^2

Table 6.1

6.2 Stationarity in variance

How to make a series stationary in variance? **Box-Cox** transform: it's a series of parametric family of transformation to stabilisze the variance of x_t

$$z_t(\lambda) = \begin{cases} \frac{x_t^\lambda - 1}{\lambda} & \lambda \neq 0 \\ \log(x_t) & \lambda = 0 \end{cases}$$

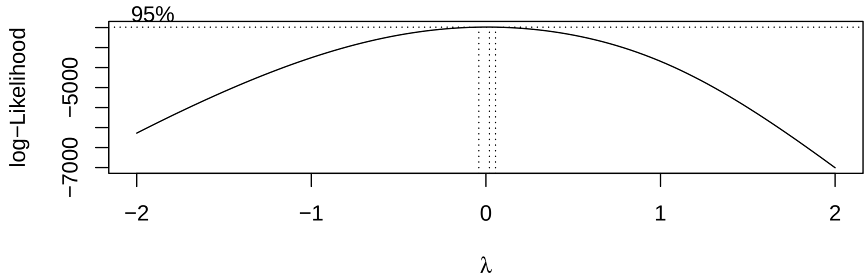
where λ is a parameter to be determined. Given the estimated parameter we apply a transformation to the series to make it stable 6.1.
In practise, it is advised to choose the value from the table rather than the precise value if the estimated transformation parameter is close to one of the values of the previous table because the value from the table is simpler to understand. For example

- $\lambda = 1 \implies \sigma$ is constant $\implies$ no need to transform
- $\lambda = 0 \implies \sigma$ is linearly proportional to μ, $\implies$ log transform is needed
- $\lambda = \frac{1}{2} \implies \sigma = k\sqrt{\mu}$ (slow variation) $\implies$

```
library(MASS)

##
## Caricamento pacchetto: 'MASS'
## Il seguente oggetto è mascherato da 'package:lbts':
##
##     deaths

set.seed(1)
x <- exp(rnorm(1000))
bc <- boxcox(x~1) # suggested lambda=0, log(x) to make it normal
```




```r
bc$x[which.max(bc$y)]

## [1] 0.0202

# non è l'intervallo di confidenza ma meglio che niente
# da https://stackoverflow.com/questions/34550744
bc$x[bc$y > max(bc$y) - 1/2 * qchisq(.95,1)]

## [1] -0.0202  0.0202

# Reference: Linear Models With R by Julian Faraway Section 8.1.
```

6.3 Box-Jenkins procedure

Iterative procedure for identification, estimation and forecasting of seasonal ARIMA models

1. **Preliminary analysis**: graphical analysis. Plot the series and check for stationarity and seasonality:

    - **stationarity** can be assessed looking at the plot. This should show constant location and scale, suggesting that mean and variance of the time series are constant. Furthermore, *non*-stationarity can also be detected from an autocorrelation plot with a very slow decay

    - **seasonality**: can usually be assessed from an **autocorrelation plot**, a **seasonal subseries** (da gennaio a dicembre si plotta la media dei diversi mesi negli anni) or a **spectral density plot**. If it exists, one need to identify the order for the seasonal autoregressive and seasonal moving average terms.
    For many series, the period is known and a single seasonality term is sufficient: for example, for monthly data one would typically include either a seasonal AR 12 term or a seasonal MA 12 term.

2. **Transformation** for making the series **stationary** in mean (detrending or differencing) and in variance (Box-Cox transform)

- **differencing**: Box and Jenkins recommend the differencing approach to achieve stationarity. However, fitting a curve and subtracting the fitted values from the original data can also be used in the context of Box–Jenkins models.
- **seasonal differencing**: For Box–Jenkins models, one does not explicitly remove seasonality before fitting the model. Instead, one includes the order of the seasonal terms in the model specification to the ARIMA estimation software. However, it may be helpful to apply a seasonal difference to the data and regenerate the autocorrelation and partial autocorrelation plots. This may help in the model identification of the non-seasonal component of the model. In some cases, the seasonal differencing may remove most or all of the seasonality effect
- box-cox transform: see above

3. **Identification**: identify p and q. Once stationarity and seasonality have been addressed, the next step is to identify the order (i.e. the p and q) of the autoregressive and moving average terms. This can be done on the basis of criteria

    - **estimated ACF and PACF**: The sample autocorrelation plot and the sample partial autocorrelation plot are compared to the theoretical behavior of these plots.
    In practice, the sample autocorrelation and partial autocorrelation functions are random variables and do not give the same picture as the theoretical functions. This makes the model identification more difficult. In particular, mixed models can be particularly difficult to identify. Although experience is helpful, developing good models using these sample plots can involve much trial and error.
    However
        - an **AR(1)** process, the **sample autocorrelation** function should have an exponentially decreasing appearance
        - **higher-order AR**: processes are often a mixture of exponentially decreasing and damped sinusoidal components. The sample autocorrelation needs to be supplemented with a **partial autocorrelation plot**. The partial autocorrelation of an AR(p) process becomes zero at lag $p+1$ and greater, so we examine the sample partial autocorrelation function to see if there is *evidence of a departure from zero*: this is usually determined by placing a 95% confidence interval on the sample partial autocorrelation plot (which is approximately $\pm 2/\sqrt{n}$ with n denoting the sample size)
        - **MA(q)**: the autocorrelation function becomes zero at lag $q+1$ and greater, so we examine the sample autocorrelation function to see where it essentially becomes zero, as above.
        The sample partial autocorrelation function is generally not helpful for identifying the order of the moving average process.

    Table 6.2 summarizes how one can use the sample autocorrelation

Shape	Indicated model
Exponential, decaying to zero	Autoregressive model. Use the partial autocorrelation plot to identify the order of the autoregressive model.
Alternating positive and negative, decaying to zero	Autoregressive model. Use the partial autocorrelation plot to help identify the order.
One or more spikes, rest are essentially zero (or close to zero)	Moving average model, order identified by where plot becomes zero.
Decay, starting after a few lags	Mixed autoregressive and moving average (ARMA) model.
All zero or close to zero	Data are essentially random.
High values at fixed intervals	Include seasonal autoregressive term.
No decay to zero (or it decays extremely slowly)	Series is not stationary.

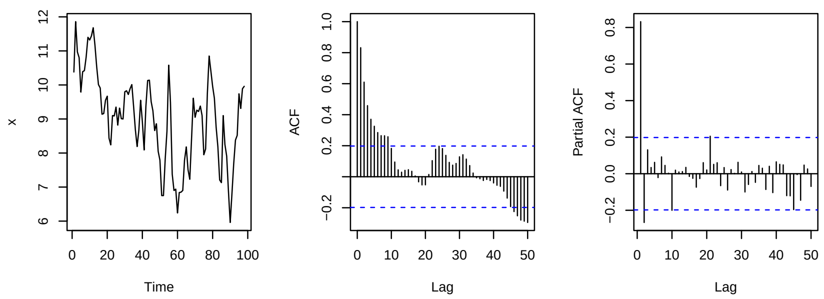
Table 6.2: Autocorrelation function shape

function for model identification.[1]

- using **information criteria** (Brockwell Davis usano come metodo primario l'AIC):

$$AIC(p, q) = \log \hat{\sigma}^2 + 2\frac{p+q}{n}$$
$$BIC(p, q) = \log \hat{\sigma}^2 \log n\frac{p+q}{n}$$

```
library(lbts)
ts_plot(lake)
```



[1] Finally, Hyndman & Athanasopoulos suggest the following

- The data may follow an ARIMA(p,d,0) model if the ACF and PACF plots of the differenced data show the following patterns:
  - the ACF is exponentially decaying or sinusoidal;
  - there is a significant spike at lag p in PACF, but none beyond lag p.
- The data may follow an ARIMA(0,d,q) model if the ACF and PACF plots of the differenced data show the following patterns:
  - the PACF is exponentially decaying or sinusoidal;
  - there is a significant spike at lag q in ACF, but none beyond lag q.

```
# https://ionides.github.io/531w20/05/notes05-annotated.pdf
aic_table <- function(data,P=5,Q=5){
  table <- matrix(NA,(P+1),(Q+1))
  for(p in 0:P) {
    for(q in 0:Q) {
      # assuming stationarity ... 0 for I component
      table[p+1,q+1] <- arima(data,order=c(p,0,q))$aic
    }
  }
  dimnames(table) <- list(
    paste("AR",0:P, sep=""),
    paste("MA",0:Q, sep=""))
  table
}

(aics <- aic_table(lake))

##       MA0   MA1   MA2   MA3   MA4   MA5
## AR0 333.3 253.3 228.9 220.1 220.5 220.7
## AR1 217.2 212.5 214.5 215.9 217.3 219.3
## AR2 213.3 214.5 216.4 217.5 218.3 220.2
## AR3 214.0 215.8 217.7 219.2 220.2 222.2
## AR4 215.6 217.2 218.4 219.8 221.3 223.0
## AR5 217.6 218.3 220.3 221.3 223.3 224.9

# lowest AIC is obtained with Arma(1,1)

# however we should also look at diagnostic plots for these models (eg look at
# residuals of estimation).

# Are there outliers:
```

4. Estimation (ML): estimating the parameters for Box-Jenkins models involves numerically approximating the solutions of nonlinear equations. Maximum likelihood estimation is generally the preferred technique

```
(lake_arma11 <- arima(lake, order=c(1,0,1)))

##
## Call:
## arima(x = lake, order = c(1, 0, 1))
##
## Coefficients:
##          ar1    ma1  intercept
##        0.745  0.321      9.055
## s.e.  0.078  0.114      0.350
##
## sigma^2 estimated as 0.475:  log likelihood = -103.2,  aic = 212.5
```

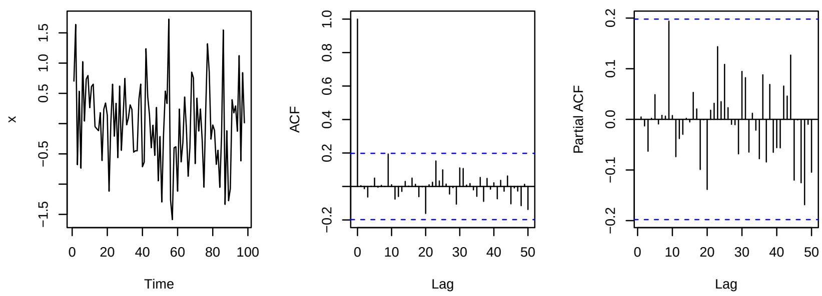
5. **Diagnostic checking**: on the estimated coefficients and on the basis of the residual analysis.

   - **estimated coefficients**: credo si riferisca a se dei coefficienti di un modello specificato non sono diversi da 0
   - **residuals**: the error term is assumed to follow the assumptions for a stationary univariate process. The residuals should be *white noise* (or independent when their distributions are normal) drawings from a fixed distribution with a constant mean and variance.
     Way to assess if the residuals from the Box–Jenkins model follow the assumptions:
     – do graphics (including an autocorrelation plot) of the residuals
     – value of the Box-Ljung statistic.

     If these assumptions are not satisfied, one needs to fit a more appropriate model. That is, go back to the model identification step and try to develop a better model. Hopefully the analysis of the residuals can provide some clues as to a more appropriate model

```
## look at residuals: they seems not autocorrelated looking at ACF and white
## noise (within the confidence interval)

ts_plot(lake_arma11$residuals)
```




```
# fuck hyndyman facciamo su tutto e ignoriamo seasonality
## boxlung test # set looking at periodicity i guess
library(stats)
sapply(1:15, function(l)
  Box.test(lake_arma11$residuals,
           type = "L", # specificarlo altrimenti fa il box-pierce
           lag=l)$p.value)

##  [1] 0.9626 0.9904 0.9342 0.9800 0.9833 0.9946 0.9983 0.9995 0.8488 0.9015 0.9054 0.921
## [13] 0.9464 0.9640 0.9780
```

6. Forecasting

If one step fails, then the procedure starts from the beginning.

Part II

Advanced time series

Chapter 7

Advanced intro

Here we see linear and non linear and non-linear time varying parameter models, unobserved component models, observation driven models. Models are *specified*, dynamic parameters recovered and static parameters estimated mainly by likelihood methods. For example in

$$\mu_{t+1} = \omega + \alpha + \mu_t + \beta(y_t - \mu_t)$$

where

- μ_{t+1} is the time varying parameter
- ω, α, β are static parameters
- μ_t is a dynamic parameter
- y_t is observation

  Prerequisites:

- Stochastic process: conditional expectation martingales difference sequences (mds)
- Time series: Linear processes (AR, MA process and random walk)
- Inference: Maximum likelihood estimation

Book:

- Durbin e Koopman (SSM and Kalman filter): for time varying means
- Francq Zakoian: for time vayring scales/variances

7.1 Review of time series

Remark 25 (Main idea of the course). If in a basic course we deal with stationary time series for which mean and variance does not change, in this course we drop this assumption focusing on time-varying mean and time-varying variances (conditional)

7.1.1 Conditional and unconditional moments of a s.p.

Considering $\{Y_t\}_{t\in T}$ as stochastic process with each single random variable being a measurable function of type $Y_t : (\Omega, \mathscr{A}, \mathbb{P}) \to (\mathbb{R}, \beta(\mathbb{R}), F_y)$ (so being measurable $Y_t^{-1}(B) \in \mathscr{A}, \forall B \in \beta(\mathbb{R})$).
The *unconditional mean* of the stochastic process

$$\mathbb{E}\left[Y_t\right] = \int_{\mathbb{R}} y \,\mathrm{d}F_y(y) = \mu_t$$

unconditional mean depend on time (when the process is stationary this is not the case, it's a constant). If $\mathscr{F}$ is the natural filtration of the process (a collection of $\mathscr{F}_t = \sigma\left\{Y_0, Y_1, \ldots, Y_t\right\}$ with $\mathscr{F}_t \subset \mathscr{F}_{t+1}$ and $\mathscr{F} = \left\{\mathscr{F}_t\right\}_{t\in T}$) we can define

- **unconditional expectation** of $Y_t$is $\mathbb{E}\left[Y_t\right] = 0$ (because $\omega = 0$)
- in **conditional** expectation we condition on some information, and in time series this is the past, constituted by the filtration $\mathbb{E}\left[Y_t|\mathscr{F}_{t-1}\right]$

Example 7.1.1 (AR(1) with $\omega = 0$)**.** Let us consider as an example, a zero mean ($\omega = 0$) stationary AR(1) process

$$Y_t = \underbrace{\omega}_{=0} + \phi Y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathrm{WN}\left(0, \sigma^2\right)$$

ω is not the expected value of the process which is $\frac{\omega}{1-\phi}$ (with $|\phi| < 1$ implying stationarity and that expectation is also finite).
By definition the WN is defined as

$$\begin{aligned}
&\mathbb{E}\left[\varepsilon_t\right] = 0\\
&\mathbb{E}\left[\varepsilon_t^2\right] = \sigma^2 < \infty\\
&\mathbb{E}\left[\varepsilon_t\varepsilon_s\right] = 0, \forall t \neq s
\end{aligned}$$

Here for unconditional[1] and conditional[2] mean we have

$$\begin{aligned}
\mathbb{E}\left[Y_t\right] &= 0\\
\mathbb{E}\left[Y_t|\mathscr{F}_{t-1}\right] &= \phi\,\mathbb{E}\left[Y_{t-1}|\mathscr{F}_{t-1}\right] + \mathbb{E}\left[\varepsilon_t|\mathscr{F}_{t-1}\right] = \phi Y_{t-1} + \mathbb{E}\left[\varepsilon_t\right] = \phi Y_{t-1}
\end{aligned}$$

Note that conditional expectation is dynamic/time varying (depends on t as index via Y_{t-1}). So we can say an *AR(1) model is a model for the conditional mean* (or same said for the time-varying mean or dynamic mean) because it gives us a time varying conditional mean of Y_t.
Unconditional[3] and conditional variance are

$$\begin{aligned}
\mathrm{Var}\left[Y_t\right] &= \frac{\sigma^2}{1-\phi^2}\\
\mathrm{Var}\left[Y_t|\mathscr{F}_{t-1}\right] &= \mathbb{E}\left[(Y_t - \mathbb{E}\left[Y_t|\mathscr{F}_{t-1}\right])^2|\mathscr{F}_{t-1}\right] = \mathbb{E}\left[\varepsilon_t^2|\mathscr{F}_{t-1}\right] = \sigma^2
\end{aligned}$$

In the latter case (going from unconditional to conditional) what does information brings in? there is an *improvement* in terms of variability (since $\phi^2 < 1$).

[1]To derive the unconditional mean of an AR(1) process either assume stationarity or use the wold representation (do it as an exercise)

[2]easier to calculate

[3]Calculated again assuming stationarity or by using wold representation

Differently from the conditional expected value, the conditional variance is not time varying/a function of the past information.
In general, in linear models $\mathrm{Var}\left[Y_t|\mathscr{F}_{t-1}\right] = \sigma^2$.
Linear models such as AR(1) are *not* good models for time-varying variance, they do not capture any time variation in the conditional variance.

Example 7.1.2 (Check)**.** Verify two useful results on conditional and unconditional expectation (by the law of iterated expectation)

$$\begin{aligned}
&\mathbb{E}\left[\mathbb{E}\left[Y_t|\mathscr{F}_{t-1}\right]\right] = \mathbb{E}\left[Y_t\right] \\
&\mathrm{Var}\left[Y_t\right] = \mathbb{E}\left[\mathrm{Var}\left[Y_t|\mathscr{F}_{t-1}\right]\right] + \mathrm{Var}\left[\mathbb{E}\left[Y_t|\mathscr{F}_{t-1}\right]\right]
\end{aligned}$$

7.2 Introduction and notation

Remark 26. Up to now we considered conditional expectation let's see ...

Definition 7.2.1 (Conditional density of sp)**.** Considering a sp $\{Y_t\}_{t\in\mathbb{Z}}$ and its natural filtration up to $\mathscr{F}_{t-1} = \sigma\{Y_{t-1}\}$ the conditional density is the parametric distribution given the information at time $t-1$

$$Y_t|\mathscr{F}_{t-1} \sim p(Y_t, \lambda_t; \theta)$$

where

- p is some density (eg Gaussian, t_ν, Cauchy which is a t with $\nu = 1$)
- p is function of y_t as any distribution, but our y_t is function of a *time varying* parameter λ_t, which characterize p.
  λ_t can be the conditional expectation $\mathbb{E}\left[Y_t|\mathscr{F}_{t-1}\right] = \lambda_t$ or conditional variance $\mathrm{Var}\left[Y_t|\mathscr{F}_{t-1}\right] = \lambda_t$
- $\theta \in \Theta \subset \mathbb{R}^m$ are static parameters which are needed to characterize the density (eg ν for a t distribution, σ for Gaussian, or constant parameters that are used to specify the evolution of λ_t)

Remark 27. We're interested in dynamics of the time varying parameter we want to see how λ_t does vary in time (thus we can say how the conditional density varyies in time).
Basically we want to recover λ_t based on observations $\{y_1, \dots, y_T\}$

Example 7.2.1. If the model is

$$Y_t|\mathscr{F}_{t-1} \sim \phi(Y_t, \mu_t; \theta) \tag{7.1}$$

where ϕ is Gaussian density with mean μ_t and static scale σ that is

$$f_{Y_t|\mathscr{F}_{t-1}}(y_t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\left(-\frac{1}{2}\left(\frac{y_t-\mu_t}{\sigma}\right)^2\right)}$$

we have that

- $\lambda_t = \mu_t$ is the time varying parameter (mean), whose evolution depends on some parameter $\mathbf{v}$. For example

  $$\lambda_{t+1} = \omega + \alpha\lambda_t + \beta(y_t - \lambda_t) \tag{7.2}$$

  where $\mathbf{v} = (\omega, \alpha, \beta)$

- $\theta = (\sigma, \mathbf{v}) \in \Theta \subset \mathbb{R}^m$ is the static parameter (in this case $m = 3 + 1$)

Important remark 17. In any of these example we need to specify the way the time varying parameter get updated as we did in eq. 7.2.

Example 7.2.2. For example in AR(1) we have

$$\lambda_t = \phi Y_{t-1}$$

assuming $\varepsilon_t \sim \text{NID}\left(0, \sigma^2\right)$. Here we have $m = 2$ and $v = \phi$.

Definition 7.2.2 (Observation driven model)**.** Both in AR(1) that in example 7.2 $\lambda_t$is function of y_{t-1} with no additional error terms. These are called **observation driven model**: the dynamics of the time varying parameter depends on past observation

Example 7.2.3. If the model is

$$Y_t | \mathscr{F}_{t-1} \sim t_\nu(Y_t, \sigma_t; \theta)$$

differently from 7.1 we have

- t_ν is the conditional distribution with constant location μ so

  $$f_{Y_t|\mathscr{F}_{t-1}} = c_\nu \left(1 + \frac{(y_t - \mu)^2}{\nu \sigma_t^2}\right)^{-\frac{\nu+1}{2}}$$

  with c_ν a constant associated to the student t density for ν degrees of freedom

- $\lambda_t = \sigma_t$ is now the dynamic/time varying parameter whose evolution depends on some set of parameters $\mathbf{v}$, eg we could have

  $$\sigma_t^2 = \omega + \alpha y_{t-1}^2$$

  Note that $\lambda_t = \sigma_t$ is not actually the variance of the distribution (which for t is $\sigma_t \frac{\nu}{\nu-2}$, with $\nu > 2$ for the variance to exists) but proportional to it

- μ is static here

- $\theta = (\mu, \nu, \mathbf{v}) \in \Theta \subset \mathbb{R}^m$ with the first two parameters characterizing the student t distribution and $\mathbf{v} = (\omega, \alpha)$ the parameters regarding the dynamic update (credo sia un arch 1 model)

Remark 28 (Model we will see). Models for time varying parameter that can be mean or variance μ_t, σ_t (or quantiles q_t or any other parameter that could characterize a parametric distribution). It doesn't have to be a moment, it can be a quantity that serves to characterize the distribution.

7.3 Decomposition

We've said that the conditional distribution of our process follows a distribution $Y_t|\mathscr{F}_{t-1} \sim p$ for some distribution p.
An equivalent representation for models with time varying mean *or* variance are the following:

$$
\begin{aligned}
Y_t &= \lambda_t + \sigma\varepsilon_t \\
Y_t &= \mu + \lambda_t\varepsilon_t
\end{aligned}
$$

with $\varepsilon_t \sim \mathrm{IID}\,(0,1)$. The

- first is the *signal plus noise decomposition* with λ_t latent unobserved signal and ε_t noise component
- second is *location-scale* representation (with μ location and λ_t scale)

Our problem is to specify a model for λ_t as a function of $\mathscr{F}_{t-1}$
If we take a general decomposition as the following

$$y_t = \mu_t + \varepsilon_t$$

where we see a signal and a noise both depending on time, by making some assumptions on the behaviour of things in time, we can encapsulate all the models we'll see. For some model on varying mean

TODO: chiari errori suoi da copia incolla

1. $\mu_t = y_{t-1}$ with $\varepsilon_t \sim \mathrm{IID}\left(0,\sigma^2\right)$ (RW)
2. $\mu_t = \phi y_{t-1}$ with $\varepsilon_t \sim \mathrm{WN}\left(0,\sigma^2\right)$ (AR(1))
3. $\mu_t = \theta\varepsilon_{t-1}$ with $\varepsilon_t \sim \mathrm{WN}\left(0,\sigma^2\right)$ (MA(1))
4. $\mu_{t+1} = \mu_t + \eta_t$ with $\eta_t \sim \mathrm{N}\left(0,\sigma^2_\eta\right)$ and $\varepsilon_t \sim \mathrm{N}\left(0,\sigma^2_\varepsilon\right)$ (LLM)
5. $\mu_{t+1} = \phi\mu_t + \eta_t$ with $\eta_t \sim \mathrm{N}\left(0,\sigma^2_\eta\right)$ and $\varepsilon_t \sim \mathrm{N}\left(0,\sigma^2_\varepsilon\right)$ (AR(1)+noise)
6. $\mu_{t+1|t} = \omega + \phi\mu_{t|t-1} + kv_t$, with $v_t = y_t - \mu_{t|t-1}$ $\eta_t \sim \mathrm{N}\left(0,\sigma^2_\eta\right)$ (IFKF)
7. $\mu_{t+1|t} = \omega + \phi\mu_{t|t-1} + u_t$ with u_t score and ε_t heavy tailed (score driven model)

For scale

1. $\mu_t = 0$, $\varepsilon_t = \sigma_t z_t$, with $z_t \sim \mathrm{IID}\,(0,1)$, $\sigma^2_t = \omega + \alpha y^2_{t-1}$ (ARCH)
2. $\mu_t = 0$, $\varepsilon_t = \sigma_t z_t$, with $z_t \sim \mathrm{IID}\,(0,1)$, $\sigma^2_t = \omega + \alpha y^2_{t-1} + \beta\sigma^2_{t-1}$ (GARCH)

Example 7.3.1 (1) random walk)**.** For the first let

$$
\begin{aligned}
y_t &= \mu_t + \varepsilon_t \\
\mu_t &= y_{t-1}
\end{aligned}
$$

with $\varepsilon_t \sim \mathrm{IID}\left(0,\sigma^2\right)$. Plugging in we obtain

$$y_t = y_{t-1} + \varepsilon_t$$

So this case is that $\{Y_t\}_{t\in T}$ is a **random walk process**.
A RW can be seen also like by putting

$$\begin{aligned} Y_0 &= 0 \\ Y_1 &= Y_0 + \varepsilon_1 = \varepsilon_1 \\ Y_2 &= Y_1 + \varepsilon_2 = \varepsilon_1 + \varepsilon_2 \\ Y_t &= \sum_{k=1}^{t} \varepsilon_k \end{aligned}$$

Here

$$\begin{aligned} \mathbb{E}\left[Y_t\right] &= 0 \\ \text{Var}\left[Y_t\right] &= t \cdot \sigma^2 \end{aligned}$$

and the latter implies that the process is not weakly stationary (variance depending on t). Finally

TODO: check sta roba non mi fido

$$\begin{aligned} \mathbb{E}\left[Y_t \cdot Y_{t+k}\right] &= t \cdot \sigma^2, \quad k > 0 \\ \mathbb{E}\left[Y_t \cdot Y_{t-k}\right] &= (t-k) \cdot \sigma^2, \quad k > 0 \end{aligned}$$

Covariance function is non symmetric (the process is not stationary so we're not surprised ...). The autocovariance function is linearly decreasing for k increasing (but never goes to 0)
These above were unconditional moments; to calculate conditional moments

$$\mathbb{E}\left[Y_t | \mathscr{F}_{t-1}\right] = \mathbb{E}\left[Y_{t-1} | \mathscr{F}_{t-1}\right] + \mathbb{E}\left[\varepsilon_t | \mathscr{F}_{t-1}\right] = Y_{t-1}$$

and we conclude that RW is a martingale process. If we take the differences

$$(1-L)Y_t = Y_t - Y_{t-1} = \varepsilon_t$$

which is a *martingale difference sequence* (IID in this case as here ε_t is IID)

Definition 7.3.1 (MDS)**.** A stochastic process X is a martingale difference sequence if its expectation with respect to the past is zero. That is X_t is an MDS if it satisfies the following two conditions for any t

$$\begin{aligned} \mathbb{E}\left[|X_t|\right] &< \infty \\ \mathbb{E}\left[X_t | F_{t-1}\right] &= 0 \end{aligned}$$

By construction, this implies that if Y_t is a martingale, then $X_t = Y_t - Y_{t-1}$ will be an MDS. Hence the name

Remark 29. The MDS is an extremely useful construct in modern probability theory because it implies much milder restrictions on the memory of the sequence than independence, yet most limit theorems that hold for an independent sequence will also hold for an MDS.

Example 7.3.2 (2) zero mean AR(1))**.** For the second let

$$\begin{aligned} y_t &= \mu_t + \varepsilon_t \\ \mu_t &= \phi y_{t-1} \end{aligned}$$

with $\varepsilon_t \sim \text{WN}\left(0, \sigma^2\right)$. Plugging in we obtain

$$Y_t = \phi Y_{t-1} + \varepsilon_t$$

As seen before it's an AR(1) with zero mean; already seen conditional vs unconditional mean and variance.
Regarding autocorrelation:

$$\begin{aligned}
\mathbb{E}\left[Y_t\right] &= 0 \\
\gamma_1 &= \mathbb{E}\left[Y_t \cdot Y_{t+1}\right] = \mathbb{E}\left[Y_t \cdot (\phi Y_t + \varepsilon_{t+1})\right] \\
&= \phi\gamma_0 + \underbrace{\mathbb{E}\left[Y_t \cdot \varepsilon_{t+1}\right]}_{=0} = \phi\gamma_0 \\
\gamma_2 &= \mathbb{E}\left[Y_t \cdot Y_{t+2}\right] = \mathbb{E}\left[Y_t \cdot (\phi Y_{t+1} + \varepsilon_{t+2})\right] = \phi\gamma_1
\end{aligned}$$

So in general, for the autocovariance

$$\gamma_k = \phi\gamma_{k-1} \iff \gamma_k = \phi^k\gamma_0$$

with $\gamma_k = \gamma_{-k}$ and $\gamma_0 = \frac{\sigma^2}{1-\phi^2}$

$$\rho_k = \phi^{|k|}$$

when plotting the autocorrelation (for $\phi > 0$) it never goes to zero but is linearly decreasing
The Wold representation of AR(1) (assuming $|\phi| < 1$, condition for existence of the inverse of the polynomial in the lag operator)

$$(1 - \phi L)Y_t = \varepsilon_t Y_t = \frac{1}{1 - \phi L}\varepsilon_t = \sum_{j=0}^{\infty} \phi^j L^j \varepsilon_t$$

Example 7.3.3. For the third let

TODO: qui fa casino col MA(1) e sto - di merda

$$\begin{aligned}
y_t &= \mu_t + \varepsilon_t \\
\mu_t &= -\theta\varepsilon_{t-1}
\end{aligned}$$

with $\varepsilon_t \sim \text{WN}\left(0, \sigma^2\right)$. Plugging in we obtain

$$y_t = \varepsilon_t - \theta\varepsilon_{t-1}$$

which is a MA(1).

$$\begin{aligned}
\mathbb{E}\left[Y_t\right] &= 0 \\
\rho_k &= \begin{cases} 1 & k = 0 \\ \frac{-\theta}{1+\theta^2} & |k| = 1 \\ 0 & |k| > 1 \end{cases}
\end{aligned}$$

Here the autocorrelation if $\theta < 0$ is decreasing and becomes zero after the q-th lag

$$
\begin{aligned}
Y_t &= (1 - \theta L)\varepsilon_t \\
\varepsilon_t &= \sum_{j=0}^{\infty} \theta^j y_{t-j} = y_t + \theta y_{t-1} + \dots \\
y_t &= -\sum_{j=1}^{\infty} \theta^j y_{t-j} + \varepsilon_t
\end{aligned}
$$

and thus

$$\mathbb{E}\left[Y_t | \mathscr{F}_{t-1}\right] = -\sum_{j=1}^{\infty} \theta^k y_{t-j}$$

which can be done iof $|\theta| < 1$ (condition of invertibility)

Remark 30. above some processes already seen in the past, below we start looking at new processes

Example 7.3.4 (Random walk + noise (LLM))**.** If we have

$$
\begin{aligned}
y_t &= \mu_t + \varepsilon_t \\
\mu_{t+1} &= \mu_t + \eta_t
\end{aligned}
$$

with

- $\eta_t \sim \text{NID}\left(0, \sigma_\eta^2\right)$
- $\varepsilon_t \sim \text{NID}\left(0, \sigma_\varepsilon^2\right)$
- $\mathbb{E}\left[\varepsilon_t \eta_s\right] = 0$, $\forall t, s$ (uncorrelated but also gaussian so independent)

As before we have signal + noise (first equation). But we have a second equation which has a novelty wrt to what seen so far (AR, MA, RW). We have that

- μ_t is a random walk
- y_t is a *random walk + noise* (this is what this process is)
- the very novelty here is that the time varying parameter μ_{t+1} depends on the term η_t, an *additional source of error* (aside from ε_t) with its own distribution and parameters

This is what is called a **parameter driven model**: the time varying parameter depends not on past value of the process
This is an Unobserved component (UC) model (there are things unobserved in the two equations), a local level model (LLM), already in state space form

Definition 7.3.2 (State space form)**.** A model written using two equations where

1. the first is observation or **measurement equation**, involving the only variable we're able to observe, y_t)

2. a **transition equation** that regulates how the unobsrved component of our interest μ_t updates itself

Definition 7.3.3 (Reduced form)**.** When we put the measurement equation and transition equation all into 1 by plugging transition equation in measurement equation

Example 7.3.5 (AR(1) + noise)**.** If we have

$$\begin{aligned} y_t &= \mu_t + \varepsilon_t \\ \mu_{t+1} &= \phi\mu_t + \eta_t \end{aligned}$$

with

- $\eta_t \sim \text{NID}\left(0, \sigma_\eta^2\right)$
- $\varepsilon_t \sim \text{NID}\left(0, \sigma_\varepsilon^2\right)$
- $\mathbb{E}\left[\varepsilon_t \eta_s\right] = 0, \forall t, s$

The only difference wrt to previous is ϕ. This is an AR(1) + noise (already in state space form), where the signal is an AR(1) process.
This again is a parameter driven model: here the *static parameters* are $\phi, \sigma_\eta^2, \sigma_\epsilon^2$.
The reduced form is an ARMA(1,1) with restriction on the parameters (we'll see this)

Remark 31. Now we see observation driven model.
With notation $\mu_{t+1|t}$ we emphasize that given the information at time t, $\mathscr{F}_t$, μ_{t+1} is known

Example 7.3.6 (IFKF)**.** The acronym is "innovation form of Kalman filter" and is

$$\begin{aligned} y_t &= \mu_t + \varepsilon_t \\ \mu_{t+1|t} &= \omega + \phi\mu_{t|t-1} + kv_t \end{aligned}$$

where $v_t = y_t - \mu_{t|t-1}$ is what is called *one step ahead innovation error*
In this specification/representation μ_t just depends on past observations

Remark 32. Now model for time varying variance

Example 7.3.7 (ARCH(1))**.**

$$\begin{aligned} y_t &= \mu_t + \varepsilon_t \\ \mu_t &= 0 \\ \varepsilon_t &= \sigma_t z_t \\ \sigma_t^2 &= \omega + \alpha y_{t-1}^2 \end{aligned}$$

so summarizing

$$\begin{aligned} y_t &= \sigma_t z_t \\ \sigma_t^2 &= \omega + \alpha y_{t-1}^2 \end{aligned}$$

with $z_t \sim \text{IID}(0,1)$.
If we calculate the conditional expectation (σ_t is F_{t-1} measurable by its definition)

$$\mathbb{E}[Y_t|\mathscr{F}_{t-1}] = \mathbb{E}[\sigma_t z_t|\mathscr{F}_{t-1}] = \sigma_t \, \mathbb{E}[z_t|\mathscr{F}_{t-1}] = 0$$

The conditional variance is time varying

$$\mathbb{E}\left[Y_t^2|\mathscr{F}_{t-1}\right] = \mathbb{E}\left[\sigma_t^2 z_t^2|\mathscr{F}_{t-1}\right] = \sigma_t^2 \, \mathbb{E}\left[z_t^2|\mathscr{F}_{t-1}\right] = \sigma_t^2 \cdot 1(= \omega + \alpha y_{t-1}^2)$$

What does this mean? How do we created a model for time varying variance? we have scaled an iid random variable z_t by multiplying it by σ_t which is $\mathscr{F}_{t-1}$-measurable (it's the square root of $\omega + \alpha y_{t-1}^2$ so it's $\mathscr{F}_{t-1}$-measurable). That implies that the original model has (one step ahead) conditional distribution

$$p(y_t|\mathscr{F}_{t-1}) \sim \text{IID}\left(0, \sigma_t^2\right)$$

with σ_t^2 time varying and depends on y_{t-1}^2.
THis is an ARCH(1) model for σ_t^2: AR stands for autoregressive and CH for conditional heteroskedasticity (conditional variance changes in time).
If we calculate the unconditional mean, by the law of iterated expectation

$$\mathbb{E}[Y_t] = \mathbb{E}[\mathbb{E}[Y_t|F_{t_1}]] = 0$$

while the unconditional variance (we will see) is

$$\text{Var}[Y_t] = \frac{\omega}{1-\alpha}$$

Example 7.3.8 (GARCH)**.** G stands for *generalized.* Summarizing it

$$\begin{aligned} y_t &= \sigma_t z_t \\ \sigma_t^2 &= \omega + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2 \end{aligned}$$

with $z_t \sim \text{IID}(0,1)$-
Again σ_t^2 is F_{t-1}-measurable and

$$\mathbb{E}\left[Y_t^2|\mathscr{F}_{t-1}\right] = \sigma_t^2$$

Here σ_t^2 has a recursion and depends on its past values and past values of the series

Chapter 8

Unobserved component models

Definition 8.0.1 (State space form)**.** A model written using two equations where

NB: reference Durbin e Koopman cap 1 e 2

1. the first is observation or **measurement equation**, involving the only variable we're able to observe, y_t)
2. a **transition equation** that regulates how the unobserved component of our interest μ_t updates itself

8.1 Local level model

Remark 33. An Unobserved component (UC) model (there are things unobserved in the two equations of the state space form), already in state space form is the following **local level model**, also called linear Gaussian model (because both the equation are linear function of Gaussian errors). Linearity can be dropped but not gaussianity for the nice properties we'll see to hold

Definition 8.1.1 (LLM, Random walk + noise)**.** If we specify the model like

$$\begin{aligned} y_t &= \mu_t + \varepsilon_t \quad \varepsilon_t \sim \text{NID}\left(0, \sigma_\varepsilon^2\right) \\ \mu_{t+1} &= \mu_t + \eta_t \quad \eta_t \sim \text{NID}\left(0, \sigma_\eta^2\right) \end{aligned}$$

with

- ε_t and η_t independent for all $t, s \in \mathbb{Z}$ ($\mathbb{E}\left[\varepsilon_t \eta_s\right] = 0$, so uncorrelated but also gaussian so independent);
- $\mu_1 \sim \text{N}\left(a_1, P\right)$ (an initialization step, can be a constant parameter actually or have some distribution with known properties)

we have that

- y_t has a *random walk + noise* shape (this is what this process is) where μ_t is a random walk

- the time varying parameter μ_{t+1} depends on its own *source of error* η_t, with its distribution and parameters, but *not on past value of the process* (eg y_t) which makes this a a **parameter driven model**

Remark 34. We now start studying properties of the model; in time series mainly means autocovariance, autocorrelation and in essence the dependence relation among variables $\{Y_t\}$, and between $\{Y_t\}$ and $\{\mu_t\}$ (here we have actually two stochastic processes).
In practice we analyze:

- the behavior at limit cases
- workout the moments and especially autocovariance function

Limit cases As special cases, if

- $\sigma_\varepsilon^2 = 0$ we obtain random walk

$$\begin{aligned} y_t &= \mu_t \\ \mu_{t+1} &= \mu_t + \eta_t \end{aligned}$$

  thus $y_t \sim RW$

- $\sigma_\eta^2 = 0$ we have a independent white noise with/plus a constant mean μ_t

$$\begin{aligned} y_t &= \mu_t + \varepsilon_t \\ \mu_{t+1} &= \mu_t = c \end{aligned}$$

  thus $y_t \sim \text{NID}\left(c, \sigma_\eta^2\right)$

Moments To obtain it we rewrite the process as a single equation (the *reduced form*) in order to apply what needed to get autocovariance. By replacing the transition equation in the measurements equation we have

$$\begin{aligned} y_{t+1} &= \mu_{t+1} + \varepsilon_{t+1} \\ &= \mu_t + \eta_t + \varepsilon_{t+1} \end{aligned}$$

from which by subtracting $y_t = \mu_t + \varepsilon_t$ from both members

$$\begin{aligned} y_{t+1} - y_t &= \mu_t + \eta_t + \varepsilon_{t+1} - \mu_t - \varepsilon_t \\ &= \eta_t + \varepsilon_{t+1} - \varepsilon_t \end{aligned}$$

So in general if $\Delta = (1 - L)$ (with L lag operator and $LX_t = X_{t-1}$ and $L^k X_t = X_{t-k}$) we can write the reduced form of a LLM (going back of 1 wrt to previous btw ...) as

$$\begin{aligned} \Delta y_t &= \eta_{t-1} + \varepsilon_t - \varepsilon_{t-1} \\ &= \varepsilon_t - \varepsilon_{t-1} + \eta_{t-1} \end{aligned}$$

So above we see that the (lagged) reduced form of a LLM is a MA(1). We can now easily calculate (unconditional) expected value, variance, covariance of lag 1

$$\begin{aligned}
\mathbb{E}\left[\Delta y_t\right] &= \mathbb{E}\left[\varepsilon_t - \varepsilon_{t-1} + \eta_{t-1}\right] = \mathbb{E}\left[\varepsilon_t\right] - \mathbb{E}\left[\varepsilon_{t-1}\right] + \mathbb{E}\left[\eta_{t-1}\right] = 0 \\
\mathbb{E}\left[(\Delta y_t)(\Delta y_t)\right] &= \mathbb{E}\left[(\varepsilon_t - \varepsilon_{t-1} + \eta_{t-1})(\varepsilon_t - \varepsilon_{t-1} + \eta_{t-1})\right] = \sigma_\varepsilon^2 + \sigma_\varepsilon^2 + \sigma_\eta^2 = 2\sigma_\varepsilon^2 + \sigma_\eta^2 \\
\mathbb{E}\left[(\Delta y_t)(\Delta y_{t-1})\right] &= \mathbb{E}\left[(\varepsilon_t - \varepsilon_{t-1} + \eta_{t-1})(\varepsilon_{t-1} - \varepsilon_{t-2} + \eta_{t-2})\right] = -\sigma_\varepsilon^2 \\
\mathbb{E}\left[(\Delta y_t)(\Delta y_{t-2})\right] &= \mathbb{E}\left[(\varepsilon_t - \varepsilon_{t-1} + \eta_{t-1})(\varepsilon_{t-2} - \varepsilon_{t-3} + \eta_{t-3})\right] = 0
\end{aligned}$$

Thus to summarize expected value and autocovariance (with variance included)

$$\begin{aligned}
\mathbb{E}\left[\Delta y_t\right] &= 0 \\
\gamma_k &= \begin{cases} 2\sigma_\varepsilon^2 + \sigma_\eta^2 & k = 0 \\ -\sigma_\varepsilon^2 & k = 1, -1 \\ 0 & |k| > 1 \end{cases}
\end{aligned}$$

The implication of the features above is that the (lagged) reduced form of a LLM is a MA(1) ($X_t = \varepsilon_t - \theta\varepsilon_{t-1}$). Thus is stationary and invertible if its underlying $|\theta| < 1$

Finally, for autocorrelation we have in general that $\rho_k = \frac{\gamma_k}{\gamma_0}$. Here:

$$\rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{-\sigma_\varepsilon^2}{2\sigma_\varepsilon^2 + \sigma_\eta^2} = -\frac{1}{\frac{\sigma_\eta^2}{\sigma_\varepsilon^2} + 2} = -\frac{1}{q+2}$$

Thus for autocorrelation, finally

$$\rho_k = \begin{cases} 1 & k = 0 \\ -\frac{1}{q+2} & k = 1, -1 \\ 0 & |k| > 1 \end{cases}$$

The $q = \frac{\sigma_\eta^2}{\sigma_\varepsilon^2}$

- is the called signal-to-noise ratio (ratio between the variance of the signal μ_t, and the variance of the noise ε_t)
- tells us how much of the overall variability that we observed in a series is explained by variance of signal or of noise.
- has range $0 < q < +\infty$ and thus $\rho_1 = -\frac{1}{q+2}$ is not in $[-1, 1]$ but in $-1/2 < \rho_1 < 0$ (for $q \to \infty$ and $q \to 0$ respectively)

Thus

- the reduced form of δy_t is a MA(1) model with $-1/2 < \rho_1 < 0$. So θ (the correlation parameter in MA(1)) we're acquainted with is not
- the reduced form of y_t is an ARIMA(0, 1, 1) model with $-1/2 < \rho_1 < 0$